PROBLEMS SHEET 2 • WORKED SOLUTIONS

Department of Mathematics • UC3M Aerospace Engineering

Chapter 2 Problem Set: First-Order ODEs and Qualitative Dynamics

Worked solutions to the 17 exercises of `ProblemsCh2.pdf`, each stated in full as in the official sheet and solved in four phases: classification and hypotheses, choice of method and change of variables, step-by-step derivation, and verification with limits. Topics: direct integration, separable equations, integrating factors, exact equations, Bernoulli substitutions, Picard uniqueness and its failure, phase lines, and finite-time blow-up.

Source Document:	sources/cuatrimestre-1/04-advanced-maths/unit-02-first-order-odes/problemas/ProblemsCh2.pdf

EXERCISE 1

Direct Integration: General Solutions

Original statement · ProblemsCh2.pdf

Direct integration • Antiderivatives

PROBLEM STATEMENT

Find the general solution to the following equations:

(i) $y'(x) = e^{3x} - x$;
(ii) $x\,y'(x) = 1$;
(iii) $y'(x) = x\,e^{x^2}$;
(iv) $(1+x)\,y'(x) = x$;
(v) $(1+x^2)\,y'(x) = x$.

Source: ProblemsCh2.pdf, Exercise 1 (page 1).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Each equation is first order, explicit once the coefficient of y' is divided out, and the unknown y does not appear on the right-hand side:

$$\frac{dy}{dx} = f(x).$$

- (i) $f(x) = e^{3x} - x$, continuous on $\mathbb{R}$.
- (ii) $f(x) = 1/x$, continuous on $(-\infty, 0)$ and on $(0, \infty)$; at $x = 0$ the equation reads $0 = 1$, so no solution can be defined at $x = 0$.
- (iii) $f(x) = xe^{x^2}$, continuous on $\mathbb{R}$.
- (iv) $f(x) = x/(1 + x)$, continuous on $(-\infty, -1)$ and on $(-1, \infty)$; at $x = -1$ the equation reads $0 = -1$, so no solution can be defined at $x = -1$.
- (v) $f(x) = x/(1 + x^2)$, continuous on $\mathbb{R}$ because $1 + x^2 \geq 1 > 0$.

Unknown: a function y differentiable on an interval I on which f is continuous. Because f does not depend on y, the right-hand side is trivially Lipschitz in y and the existence-uniqueness theorem (Robinson, Theorem 6.2) gives exactly one solution for each initial condition $y(x_0) = y_0$ on every such interval.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: when $y' = f(x)$ with f continuous on an interval, the Fundamental Theorem of Calculus gives an antiderivative F, and two antiderivatives on an interval differ by a constant (a function with zero derivative on an interval is constant, by the Mean Value Theorem). So direct integration produces all solutions.

Change of variables: none is needed for the equation itself. Where an integral is not immediate, a substitution is written explicitly: $u = 3x$ in (i), $u = x^2$ in (iii), $u = 1 + x^2$ in (v), and a polynomial division (rewriting $x = (1 + x) - 1$) in (iv). Constants of integration are attached separately on each interval of continuity.

$$y(x) = \int f(x)\,dx = F(x) + C.$$

METHOD OF THE CLASS NOTES [NOTES, PP. 2-3]

The class notes integrate a first-order equation from the initial point to the current point and close with Barrow's rule (the population equation, p. 2; Step 2 of the integrating-factor algorithm, p. 3). When the right side does not contain y, $y' = f(x)$ with f continuous on an interval J, pick any $x_0 \in J$:

$$\int_{x_0}^{x} y'(s)\,ds = y(x) - y(x_0) = \int_{x_0}^{x} f(s)\,ds,$$

so the free value $y(x_0)$ is the constant C of the general solution, one for each interval of continuity. In the form (L) of the notes, (ii), (iv) and (v) read $a_0(x)\,y' = b(x)$; dividing by a_0 is allowed only where $a_0 \neq 0$, which is why (ii) and (iv) split into two intervals.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): $y' = e^{3x} - x$

Integrate term by term:

$$y(x) = \int e^{3x}\,dx - \int x\,dx.$$

For the first integral put $u = 3x$, so $du = 3\,dx$ and $dx = du/3$:

$$\int e^{3x}\,dx = \frac{1}{3}\int e^{u}\,du = \frac{1}{3}e^{u} = \frac{1}{3}e^{3x}.$$

The second integral is $\int x\,dx = x^2/2$. Hence

$$y(x) = \frac{1}{3}e^{3x} - \frac{x^2}{2} + C, \qquad C \in \mathbb{R},\ x \in \mathbb{R}.$$

Part (ii): $x\,y' = 1$

On each interval $x \neq 0$ divide by x to get $y' = 1/x$. Then

$$y(x) = \int \frac{dx}{x} = \ln|x| + C.$$

The absolute value is needed: for $x > 0$, $\frac{d}{dx}\ln x = 1/x$; for $x < 0$, $\frac{d}{dx}\ln(-x) = \frac{-1}{-x} = \frac{1}{x}$. The two half-lines are not connected, so the constants are independent:

$$y(x) = \begin{cases} \ln x + C_1, & x > 0, \\ \ln(-x) + C_2, & x < 0, \end{cases} \qquad C_1, C_2 \in \mathbb{R}.$$

Part (iii): $y' = xe^{x^2}$

Substitute $u = x^2$, so $du = 2x\,dx$ and $x\,dx = du/2$:

$$y(x) = \int xe^{x^2}\,dx = \frac{1}{2}\int e^{u}\,du = \frac{1}{2}e^{u} + C = \frac{1}{2}e^{x^2} + C.$$

Part (iv): $(1 + x)\,y' = x$

On each interval $x \neq -1$ divide by $1 + x$ and rewrite the numerator as $x = (1 + x) - 1$:

$$y'(x) = \frac{x}{1 + x} = \frac{(1 + x) - 1}{1 + x} = 1 - \frac{1}{1 + x}.$$

Integrate, using $u = 1 + x$, $du = dx$ for the second term:

$$y(x) = \int 1\,dx - \int \frac{du}{u} = x - \ln|1 + x| + C.$$

Again the constants are independent on $(-\infty, -1)$ and on $(-1, \infty)$:

$$y(x) = x - \ln|1 + x| + C_{\pm}.$$

Part (v): $(1 + x^2)\,y' = x$

Since $1 + x^2 > 0$ on all of $\mathbb{R}$, divide: $y' = x/(1 + x^2)$. Substitute $u = 1 + x^2$, so $du = 2x\,dx$ and $x\,dx = du/2$:

$$y(x) = \int \frac{x\,dx}{1+x^2} = \frac{1}{2}\int \frac{du}{u} = \frac{1}{2}\ln|u| + C = \frac{1}{2}\ln(1+x^2) + C.$$

No absolute value is needed because $u = 1 + x^2 > 0$.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check each result by differentiating (chain rule written out):

- (i) $\frac{d}{dx}\left[\frac{1}{3}e^{3x} - \frac{x^2}{2} + C\right] = \frac{1}{3}\cdot 3e^{3x} - x = e^{3x} - x.$
- (ii) $x \cdot \frac{d}{dx}\ln|x| = x \cdot \frac{1}{x} = 1$ for $x \neq 0$.
- (iii) $\frac{d}{dx}\left[\frac{1}{2}e^{x^2}\right] = \frac{1}{2}e^{x^2}\cdot 2x = xe^{x^2}.$
- (iv) $(1+x)\,\frac{d}{dx}\left[x - \ln|1+x|\right] = (1+x)\left(1 - \frac{1}{1+x}\right) = (1+x) - 1 = x.$
- (v) $(1+x^2)\,\frac{d}{dx}\left[\frac{1}{2}\ln(1+x^2)\right] = (1+x^2)\cdot\frac{1}{2}\cdot\frac{2x}{1+x^2} = x.$

Symbolic check with sympy (dsolve and differentiation of the closed forms) reproduces the five results above.

Domain remarks: the solutions of (i), (iii), (v) are defined on all of $\mathbb{R}$. For (ii) and (iv) the solution cannot be extended through $x = 0$ and $x = -1$ respectively, because the equation itself is inconsistent there ($0 = 1$ and $0 = -1$), and $\ln|x|$, $\ln|1+x|$ are unbounded at those points. The equations in (iv) and (v) look similar but behave differently: only (iv) has a singular point, since $1 + x$ vanishes at $x = -1$ whereas $1 + x^2$ never vanishes.

RESULTS

Part	General solution	Domain
(i)	$y = \frac{1}{3}e^{3x} - \frac{1}{2}x^2 + C$	$\mathbb{R}$
(ii)	$y = \ln\|x\| + C_{\pm}$	$x \neq 0$
(iii)	$y = \frac{1}{2}e^{x^2} + C$	$\mathbb{R}$
(iv)	$y = x - \ln\|1+x\| + C_{\pm}$	$x \neq -1$
(v)	$y = \frac{1}{2}\ln(1+x^2) + C$	$\mathbb{R}$

EXERCISE 2

Separable Equations and Asymptotic Integrals

Original statement · ProblemsCh2.pdf

Separation of variables • Blow-up

PROBLEM STATEMENT

Solve the following equations:

(i) $\dot{x} = t^3(1-x)$ with $x(0) = 3$;
(ii) $y' = (1+y^2)\tan x$ with $y(0) = 1$;
(iii) $\dot{x} = t^2 x$ (general solution);
(iv) $\dot{x} = -x^2$ (general solution);
(v) $dy/dt = e^{-t^2} y^2$: give the solution in terms of an integral and describe the behavior as $t \to \infty$ depending on the initial condition $y(0)$. Use that $\int_0^\infty e^{-s^2}\, ds = \sqrt{\pi}/2$.

Source: ProblemsCh2.pdf, Exercise 2 (page 1).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

All five equations have the separable form $\dot{x} = g(t)\,h(x)$ (the letters $x(t), y(x)$ or $y(t)$ are used as in the statement).

- (i) $g(t) = t^3, h(x) = 1 - x$. $f(t,x) = t^3(1-x)$ and $\partial f/\partial x = -t^3$ are continuous on $\mathbb{R}^2$, so the IVP has a unique local solution (Robinson, Theorem 6.2). The only equilibrium is $x \equiv 1$; since $x(0) = 3 \neq 1$ and solutions cannot cross an equilibrium (uniqueness), $x(t) > 1$ for all t.
- (ii) $g(x) = \tan x, h(y) = 1 + y^2$. f and $\partial f/\partial y = 2y\tan x$ are continuous for $x \in (-\pi/2, \pi/2)$, which contains $x = 0$. There are no equilibria because $1 + y^2 \geq 1$.
- (iii) $g(t) = t^2, h(x) = x$: linear and homogeneous; $f, \partial f/\partial x = t^2$ continuous on $\mathbb{R}^2$. Equilibrium $x \equiv 0$.
- (iv) $h(x) = -x^2$: $f, \partial f/\partial x = -2x$ continuous on $\mathbb{R}^2$. Equilibrium $x \equiv 0$. Because f grows quadratically, finite-time blow-up is possible.
- (v) $g(t) = e^{-t^2}, h(y) = y^2$: $f, \partial f/\partial y = 2ye^{-t^2}$ continuous on $\mathbb{R}^2$. Equilibrium $y \equiv 0$. Again h is quadratic, so blow-up in finite time must be checked.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: if $\dot{x} = g(t)h(x)$ and $h(x_0) \neq 0$, then near the initial point $\frac{1}{h(x)}\dot{x} = g(t)$, and integrating both sides in t (the left side by the substitution $x = x(t)$, $dx = \dot{x}\,dt$) gives $H(x(t)) - H(x_0) = G(t) - G(t_0)$, with $H' = 1/h$ and $G' = g$. This requires $h \neq 0$ along the solution (equilibria $h(x^*) = 0$ are treated separately and are excluded from the division) and g, $1/h$ continuous (Robinson, Section 8.4).

Procedure used throughout, written explicitly each time:

- find the equilibria $h(x^*) = 0$ and the interval containing x_0 where $h \neq 0$;
- separate: $\frac{dx}{h(x)} = g(t)\,dt$;
- integrate with Barrow's rule from (t_0, x_0): $[H(s)]_{x_0}^{x} = [G(\tau)]_{t_0}^{t}$, so the constant is fixed automatically;
- solve for x, track the sign inside absolute values, and determine the maximal interval of existence.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The notes write the separable equation as $x'(t) = f(x(t))\,g(t)$, $x(t_0) = x_0$; their f is the h of this problem. If $f \neq 0$ (never), divide by f and integrate from t_0 to t; with $u = x(\tau)$, $du = x'(\tau)\,d\tau$, the limits $\tau = t_0, t$ become $u = x_0, x(t)$:

$$\int_{t_0}^{t} \frac{x'(\tau)}{f(x(\tau))}\,d\tau = \int_{x_0}^{x(t)} \frac{du}{f(u)} = \int_{t_0}^{t} g(\tau)\,d\tau.$$

'Never' is guaranteed by Picard's theorem where f and f' are continuous: a solution that reached a zero x^* of f at some t_1 would solve the same IVP as the constant $x \equiv x^*$, so it would be that constant. This is the procedure listed above; its Barrow form $[H]_{x_0}^{x} = [G]_{t_0}^{t}$ fixes the constant from the initial data, as in the logistic example of the notes.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): $\dot{x} = t^3(1 - x)$, $x(0) = 3$

Equilibrium: $1 - x = 0$, i.e. $x \equiv 1$. The initial value $x(0) = 3 > 1$, so $1 - x < 0$ along the solution. Separate and integrate from $(0, 3)$ with Barrow's rule:

$$\int_{3}^{x} \frac{ds}{1 - s} = \int_{0}^{t} \tau^3\,d\tau.$$

Left side: $\int \frac{ds}{1-s} = -\ln|1 - s|$ (put $u = 1 - s$, $du = -ds$). Right side: $\left[\tau^4/4\right]_0^t = t^4/4$. Substitute the limits:

$$\Big[-\ln|1-s|\Big]_3^x = -\ln|1-x| + \ln|1-3| = -\ln|1-x| + \ln 2 = \frac{t^4}{4}.$$

Solve: $\ln|1-x| = \ln 2 - t^4/4$, hence $|1-x| = 2e^{-t^4/4}$. Because $x - 1 > 0$ for all t (no crossing of the equilibrium), $|1-x| = x - 1$ and

$$x(t) = 1 + 2e^{-t^4/4}, \qquad t \in \mathbb{R}.$$

Part (ii): $y' = (1+y^2)\tan x$**,** $y(0) = 1$

No equilibria. Separate: $\frac{dy}{1+y^2} = \tan x\, dx$. Integrate from $(0, 1)$:

$$\int_1^y \frac{ds}{1+s^2} = \int_0^x \tan\sigma\, d\sigma.$$

Left: $\arctan s$. Right: $\int \tan\sigma\, d\sigma = \int \frac{\sin\sigma}{\cos\sigma}\, d\sigma$; put $u = \cos\sigma$, $du = -\sin\sigma\, d\sigma$, giving $-\int du/u = -\ln|\cos\sigma|$. On $(-\pi/2, \pi/2)$, $\cos\sigma > 0$. Substitute the limits:

$$\arctan y - \arctan 1 = -\ln(\cos x) + \ln(\cos 0) = -\ln(\cos x).$$

With $\arctan 1 = \pi/4$ and $\ln\cos 0 = 0$:

$$\arctan y = \frac{\pi}{4} - \ln(\cos x) \quad \Longrightarrow \quad y(x) = \tan\left(\frac{\pi}{4} - \ln\cos x\right).$$

Domain: $\arctan y \in (-\pi/2, \pi/2)$ is required. Since $\cos x \le 1$, $-\ln\cos x \ge 0$, so the argument $\pi/4 - \ln\cos x \ge \pi/4 > -\pi/2$ and the constraint is $\frac{\pi}{4} - \ln\cos x < \frac{\pi}{2}$, i.e. $\ln\cos x > -\frac{\pi}{4}$, i.e. $\cos x > e^{-\pi/4}$. This defines the symmetric interval $|x| < x^*$ with

$$x^* = \arccos\left(e^{-\pi/4}\right) \approx 1.0974 \; (< \pi/2 \approx 1.5708).$$

As $x \to x^{*-}$, $\arctan y \to \pi/2$, so $y \to +\infty$: the solution blows up at the finite point x^* (and symmetrically at $-x^*$, because cos is even). The maximal interval of existence is $(-x^*, x^*)$, strictly smaller than the interval $(-\pi/2, \pi/2)$ where f is continuous. This is the situation described in Robinson, Section 6.3.

Part (iii): $\dot{x} = t^2 x$ (general solution)

Equilibrium $x \equiv 0$. For $x \neq 0$ separate and integrate (indefinite form, then fix the constant):

$$\int \frac{dx}{x} = \int t^2\, dt \quad \Longrightarrow \quad \ln|x| = \frac{t^3}{3} + c.$$

Exponentiate: $|x| = e^c e^{t^3/3}$, so $x = \pm e^c e^{t^3/3}$. Put $A = \pm e^c \in \mathbb{R} \setminus \{0\}$. The equilibrium $x \equiv 0$ corresponds to $A = 0$ and indeed satisfies the equation, so

$$x(t) = A\, e^{t^3/3}, \qquad A \in \mathbb{R},\ t \in \mathbb{R}.$$

Part (iv): $\dot{x} = -x^2$ (general solution)

Equilibrium $x \equiv 0$. For $x \neq 0$, separate: $\frac{dx}{x^2} = -dt$. Integrate:

$$\int x^{-2}\, dx = -\int dt \quad \Longrightarrow \quad -\frac{1}{x} = -t + c \quad \Longrightarrow \quad x(t) = \frac{1}{t + k}, \quad k = -c.$$

The family $x = 1/(t+k)$, $k \in \mathbb{R}$, together with the equilibrium $x \equiv 0$ (which is not obtained for any finite k), is the full set of solutions. With the initial condition $x(0) = x_0 \neq 0$ one gets $k = 1/x_0$, i.e.

$$x(t) = \frac{x_0}{1 + x_0 t}.$$

Each member with $x_0 \neq 0$ blows up at $t^* = -1/x_0$. If $x_0 > 0$ the solution exists for $t > -1/x_0$ (it decays to 0 as $t \to +\infty$). If $x_0 < 0$ it exists for $t < -1/x_0 = 1/|x_0|$ and blows up forward in time at $t^* = 1/|x_0|$.

Part (v): $dy/dt = e^{-t^2} y^2$

Equilibrium $y \equiv 0$ (initial condition $y(0) = 0$). Suppose $y_0 = y(0) \neq 0$. Separate and integrate from $(0, y_0)$:

$$\int_{y_0}^{y} \frac{ds}{s^2} = \int_0^t e^{-\tau^2}\, d\tau =: I(t).$$

Left side: $[-1/s]_{y_0}^{y} = -\frac{1}{y} + \frac{1}{y_0}$. So

$$\frac{1}{y} = \frac{1}{y_0} - I(t) \quad \Longrightarrow \quad y(t) = \frac{y_0}{1 - y_0\, I(t)}, \qquad I(t) = \int_0^t e^{-\tau^2}\, d\tau.$$

This formula also gives $y \equiv 0$ for $y_0 = 0$. The function I is odd, strictly increasing ($I' = e^{-t^2} > 0$), with $I(0) = 0$ and, by the given integral, $I(t) \to \sqrt{\pi}/2$ as $t \to +\infty$. Hence on $t \geq 0$ we have $0 \leq I(t) < \sqrt{\pi}/2$. The denominator $D(t) = 1 - y_0 I(t)$ decides the behavior.

Case $y_0 \leq 0$: then $-y_0 I(t) \geq 0$, so $D(t) \geq 1 > 0$ for all $t \geq 0$ and no blow-up occurs. As $t \to \infty$:

$$y(t) \to \frac{y_0}{1 - y_0\sqrt{\pi}/2} = \frac{2y_0}{2 - \sqrt{\pi}\, y_0} =: y_\infty \leq 0.$$

Case $0 < y_0 < 2/\sqrt{\pi}$: $y_0 I(t) < y_0\sqrt{\pi}/2 < 1$, so $D(t) > 1 - y_0\sqrt{\pi}/2 > 0$ and again $y(t) \to y_\infty = \frac{2y_0}{2 - \sqrt{\pi}\, y_0} > 0$, a finite limit larger than y_0.

Case $y_0 = 2/\sqrt{\pi}$: $D(t) = \frac{2}{\sqrt{\pi}}\left(\frac{\sqrt{\pi}}{2} - I(t)\right) = \frac{2}{\sqrt{\pi}}\int_t^\infty e^{-s^2}\, ds > 0$ for every finite t but $D(t) \to 0^+$. Thus $y(t) \to +\infty$ as $t \to \infty$, without finite-time blow-up (it grows like e^{t^2} up to a power of t, since $\int_t^\infty e^{-s^2} ds \sim e^{-t^2}/(2t)$).

Case $y_0 > 2/\sqrt{\pi}$: $D(0) = 1$ and D decreases continuously to $1 - y_0\sqrt{\pi}/2 < 0$, so there is a unique $T^* > 0$ with $I(T^*) = 1/y_0$, where $D(T^*) = 0$ and $y \to +\infty$. This is finite-time blow-up. In terms of the error function, $I(t) = \frac{\sqrt{\pi}}{2}\operatorname{erf}(t)$, so $T^* = \operatorname{erf}^{-1}\left(\frac{2}{\sqrt{\pi}\, y_0}\right)$. The solution exists only on $0 \leq t < T^*$; the limit $t \to \infty$ is not reached. Example: for $y_0 = 2$, solving $I(T^*) = 1/2$ numerically (mpmath quadrature and root finding) gives $T^* \approx 0.5510$, equal to $\operatorname{erf}^{-1}(1/\sqrt{\pi})$.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Substitution checks:

- (i) $\dot{x} = 2e^{-t^4/4} \cdot \left(-\frac{4t^3}{4}\right) = -2t^3 e^{-t^4/4}$ and $t^3(1 - x) = t^3\left(-2e^{-t^4/4}\right)$. They agree, and $x(0) = 1 + 2 = 3$. As $t \to \pm\infty$, $x \to 1$ (the equilibrium), approached from above.
- (ii) With $\theta = \pi/4 - \ln\cos x$, $y = \tan\theta$, $y' = (1 + \tan^2\theta)\,\theta'$ and $\theta' = \tan x$. Hence $y' = (1 + y^2)\tan x$. At $x = 0$: $\theta = \pi/4$, $y = 1$. Numerically $x^* = \arccos(e^{-\pi/4}) \approx 1.0974$; at $x = x^* - 10^{-6}$ the formula gives $y \approx 5.1 \times 10^5$, confirming the blow-up.
- (iii) $\dot{x} = A\, t^2 e^{t^3/3} = t^2 x$.
- (iv) $\dot{x} = -\frac{1}{(t+k)^2} = -x^2$.
- (v) $y' = \frac{y_0 \cdot y_0 I'(t)}{(1 - y_0 I)^2} = y^2 e^{-t^2}$, since $I'(t) = e^{-t^2}$.

Time scales: in (ii) the natural variable is the spatial coordinate x, with blow-up at $x^* \approx 1.0974$ rather than as $x \to \pi/2$; in (iv) the blow-up time is $-1/x_0$; in (v) the blow-up time is $T^* = \operatorname{erf}^{-1}(2/(\sqrt{\pi}\, y_0))$ when $y_0 > 2/\sqrt{\pi}$, so for those data a statement about $t \to \infty$ is meaningless. For all other y_0 the long-time behavior is governed by the finite integral $\int_0^\infty e^{-s^2} ds = \sqrt{\pi}/2$.

SUMMARY OF (V) FOR $t \to \infty$

Initial condition	**Behavior**
$y_0 < 2/\sqrt{\pi}$	bounded, $y \to \frac{2y_0}{2-\sqrt{\pi}\,y_0}$ (equals 0 for $y_0 = 0$)
$y_0 = 2/\sqrt{\pi}$	$y \to +\infty$ as $t \to \infty$, no finite blow-up
$y_0 > 2/\sqrt{\pi}$	blow-up at finite T^* with $\operatorname{erf}(T^*) = 2/(\sqrt{\pi}\,y_0)$

EXERCISE 3

Integrating Factor Method and Asymptotics

Original statement · ProblemsCh2.pdf

Linear first-order · Integrating factor

PROBLEM STATEMENT

Use an integrating factor to solve the following differential equations:

(i) $\dfrac{dy}{dx} + \dfrac{y}{x} = x^2$ (find the general solution and the only solution that is finite when $x = 0$),
(ii) $\dfrac{dx}{dt} + tx = 4t$ (find the solution with $x(0) = 2$),
(iii) $\dfrac{dz}{dy} = z\tan y + \sin y$ (find the general solution),
(iv) $y' + e^{-x}y = 1$ (find the solution when $y(0) = e$, leaving your answer as an integral),
(v) $\dot{x} + x\tanh t = 3$ (find the general solution, and compare it to that for $\dot{x} + x = 3$),
(vi) $y' + 2y\cot x = 5$ (find the solution with $y(\pi/2) = 1$),
(vii) $\dfrac{dx}{dt} + 5x = t$ (find the general solution),
(viii) with $a > 0$, find the solution of the equation $\dfrac{dx}{dt} + \left[a + \dfrac{1}{t}\right]x = b$ for a general initial condition $x(1) = x_0$, and show that $x(t) \to b/a$ as $t \to \infty$.

Source: ProblemsCh2.pdf, Exercise 3 (page 1).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

All eight equations are linear first order, so they have the standard form

$$\frac{dx}{dt} + p(t)\,x = q(t),$$

with the coefficient of the derivative equal to 1. If p and q are continuous on an interval I, the right-hand side $f(t,x) = q(t) - p(t)x$ and $\partial f/\partial x = -p(t)$ are continuous, so there is exactly one solution through each (t_0, x_0), and linearity makes it exist on all of I.

Part	p	q	Interval of continuity used
(i)	$1/x$	x^2	$x>0$ and $x<0$ separately
(ii)	t	$4t$	$\mathbb{R}$
(iii)	$-\tan y$	$\sin y$	$y \in (-\pi/2 + k\pi,\ \pi/2 + k\pi)$
(iv)	e^{-x}	1	$\mathbb{R}$
(v)	$\tanh t$	3	$\mathbb{R}$
(vi)	$2\cot x$	5	$(0,\pi)$, which contains $\pi/2$
(vii)	5	t	$\mathbb{R}$
(viii)	$a + 1/t$	b	$t>0$, which contains $t=1$

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: for a linear equation $\dot{x} + px = q$ with p continuous, choose $\mu(t) = \exp\left(\int p(t)\,dt\right)$. Since $\mu' = p\mu$, the product rule gives $(\mu x)' = \mu\dot{x} + p\mu x = \mu(\dot{x} + px) = \mu q$, so the left side is an exact derivative and the equation can be integrated directly. Only continuity of p and q is needed (Robinson, Section 9.2).

Procedure: (1) read off p and q; (2) compute $P(t) = \int p\,dt$ explicitly, any antiderivative will do, and set $\mu = e^{P}$; (3) write $(\mu x)' = \mu q$; (4) integrate, using Barrow's rule $\left[\mu x\right]_{t_0}^{t} = \int_{t_0}^{t} \mu q\,ds$ when an initial condition is given; (5) divide by μ.

Formula: $x(t) = \dfrac{1}{\mu(t)}\left[\mu(t_0)x_0 + \displaystyle\int_{t_0}^{t} \mu(s)q(s)\,ds\right]$.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The notes derive the factor instead of postulating it: multiply $x' + p\,x = q$ by $I(t)$ and ask that $(Ix)' = Ix' + Ipx$; the product rule $(Ix)' = I'x + Ix'$ leaves $I' = Ip$, so $I'/I = p$, $\ln I = P$ and $I = e^{P}$ (the notes' I is the μ of this page). For an IVP they fix the antiderivative as $P(t) = \int_{t_0}^{t} p(\tau)\,d\tau$, so that $I(t_0) = e^{0} = 1$, and work in three steps: 1. P and I; 2. $\int_{t_0}^{t} (Ix)'\,d\tau = \int_{t_0}^{t} Iq\,d\tau$; 3. Barrow's rule on the left:

$$x(t) = \frac{x_0}{I(t)} + \frac{1}{I(t)}\int_{t_0}^{t} I(\tau)\,q(\tau)\,d\tau.$$

This is the formula above with $\mu(t_0) = 1$. Parts (ii), (iv) and (vi) already satisfy $\mu(t_0) = 1$, and part (viii) is written below with P taken from $t_0 = 1$. In the parts without an initial condition any antiderivative is used, as in the general solution $x = \frac{1}{I}\int Iq\,dt$ of the notes.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): $y' + y/x = x^2$

Here $p = 1/x$, $q = x^2$. An antiderivative of $1/x$ is $\ln|x|$, so $\mu = e^{\ln|x|} = |x|$. Any nonzero constant multiple of μ also works, and on each half-line $\mu = x$ or $\mu = -x$ are both valid; we use $\mu = x$ on both half-lines (check: $(xy)' = xy' + y = x(y' + y/x)$ for any $x \neq 0$). Multiply the equation by x:

$$\frac{d}{dx}\left(x\,y\right) = x \cdot x^2 = x^3.$$

Integrate: $x\,y = \int x^3\,dx = \dfrac{x^4}{4} + C$. Divide by x:

$$y(x) = \frac{x^3}{4} + \frac{C}{x}, \qquad x \neq 0.$$

Finiteness at $x = 0$: $x^3/4 \to 0$, whereas $C/x \to \pm\infty$ as $x \to 0$ unless $C = 0$. So the only solution that stays finite at $x = 0$ is

$$y(x) = \frac{x^3}{4},$$

which is smooth on all of $\mathbb{R}$ and satisfies $xy' + y = x \cdot \frac{3x^2}{4} + \frac{x^3}{4} = x^3$ also at $x = 0$.

Part (ii): $\dot{x} + tx = 4t,\ x(0) = 2$

$p = t, q = 4t$. $P(t) = \int t\,dt = t^2/2$, so $\mu = e^{t^2/2}$ and $\mu(0) = 1$. Then $(\mu x)' = 4t\,e^{t^2/2}$. Integrate from 0 to t (Barrow):

$$x(t)e^{t^2/2} - x(0)\cdot 1 = \int_0^t 4s\,e^{s^2/2}\,ds.$$

Substitute $u = s^2/2$, $du = s\,ds$; the limits $s = 0, t$ become $u = 0, t^2/2$:

$$\int_0^t 4s\,e^{s^2/2}\,ds = 4\int_0^{t^2/2} e^u\,du = 4\left[e^u\right]_0^{t^2/2} = 4\left(e^{t^2/2} - 1\right).$$

So $x(t)e^{t^2/2} = 2 + 4e^{t^2/2} - 4$, and dividing by $e^{t^2/2}$:

$$x(t) = 4 - 2e^{-t^2/2}.$$

Part (iii): $dz/dy = z\tan y + \sin y$

Rewrite in standard form with independent variable y: $z' - (\tan y)\, z = \sin y$, so $p = -\tan y$, $q = \sin y$. Compute P with $u = \cos y$, $du = -\sin y\, dy$:

$$P(y) = -\int \tan y\, dy = -\int \frac{\sin y}{\cos y}\, dy = \int \frac{du}{u} = \ln|\cos y|.$$

So $\mu = |\cos y|$; on an interval where $\cos y$ has constant sign we may use $\mu = \cos y$ (check: $(z\cos y)' = z'\cos y - z\sin y = \cos y\,(z' - z\tan y)$). Then

$$\frac{d}{dy}\,(z\cos y) = \cos y\, \sin y.$$

Integrate with $v = \sin y$, $dv = \cos y\, dy$: $\int \sin y \cos y\, dy = \int v\, dv = \tfrac{1}{2}\sin^2 y$. Hence $z\cos y = \tfrac{1}{2}\sin^2 y + C$ and

$$z(y) = \frac{\sin^2 y}{2\cos y} + \frac{C}{\cos y} = \frac{1}{2}\sin y\tan y + C\sec y, \qquad \cos y \neq 0.$$

Part (iv): $y' + e^{-x}y = 1,\ y(0) = e$

$p = e^{-x}$, $q = 1$. Choose the antiderivative that vanishes at the initial point:

$$P(x) = \int_0^x e^{-s}\, ds = \left[-e^{-s}\right]_0^x = 1 - e^{-x}, \qquad \mu(x) = e^{1-e^{-x}}, \quad \mu(0) = e^0 = 1.$$

Then $(\mu y)' = \mu$. Barrow from 0 to x:

$$y(x)\,\mu(x) - y(0)\,\mu(0) = \int_0^x e^{1-e^{-s}}\, ds = e\int_0^x e^{-e^{-s}}\, ds.$$

With $y(0) = e$ and $\mu(0) = 1$ the term $y(0)\mu(0) = e$ moves to the right-hand side. Divide by $\mu(x) = e\cdot e^{-e^{-x}}$, i.e. multiply by $1/\mu(x) = e^{-1}e^{e^{-x}}$:

$$y(x) = e^{-1}e^{e^{-x}}\left[e + e\int_0^x e^{-e^{-s}}\,ds\right] = e^{e^{-x}}\left[1 + \int_0^x e^{-e^{-s}}\,ds\right].$$

The remaining integral has no elementary antiderivative (the substitution $w = e^{-s}$ turns it into $-\int e^{-w}\,dw/w$, an exponential-integral function), so the answer is left as an integral, as requested.

Part (v): $\dot{x} + x\tanh t = 3$

$p = \tanh t = \sinh t/\cosh t$, $q = 3$. With $u = \cosh t$, $du = \sinh t\,dt$:

$$P(t) = \int \frac{\sinh t}{\cosh t}\,dt = \int \frac{du}{u} = \ln\cosh t \quad (\cosh t > 0), \qquad \mu = \cosh t.$$

$$\frac{d}{dt}\left(x\cosh t\right) = 3\cosh t \implies x\cosh t = 3\sinh t + C \implies x(t) = 3\tanh t + C\,\operatorname{sech} t.$$

Comparison equation $\dot{x} + x = 3$: $p = 1$, $\mu = e^t$, $(xe^t)' = 3e^t$, so $xe^t = 3e^t + C_2$ and $x_c(t) = 3 + C_2e^{-t}$.

- As $t \to +\infty$: $\tanh t \to 1$ and $\operatorname{sech} t \to 0$, so $x \to 3$, exactly as $x_c \to 3$. Both approach 3 at rate e^{-t} (since $\operatorname{sech} t \sim 2e^{-t}$ and $3 - 3\tanh t \sim 6e^{-2t}$).
- As $t \to -\infty$: $\tanh t \to -1$, $\operatorname{sech} t \to 0$, so $x \to -3$ for every C, whereas $x_c = 3 + C_2e^{-t}$ diverges to $\operatorname{sgn}(C_2)\cdot\infty$ unless $C_2 = 0$.

The difference comes from the coefficient: $\tanh t \to \pm 1$ changes sign with t, so the equation behaves like $\dot{x} + x = 3$ for $t \gg 1$ and like $\dot{x} - x = 3$ for $t \ll -1$, whose equilibrium is $x = -3$.

Part (vi): $y' + 2y\cot x = 5,\ y(\pi/2) = 1$

$p = 2\cot x$, $q = 5$, on $(0, \pi)$ where $\sin x > 0$. With $u = \sin x$, $du = \cos x\,dx$:

$$P(x) = 2\int \frac{\cos x}{\sin x}\,dx = 2\ln(\sin x), \qquad \mu = e^{2\ln\sin x} = \sin^2 x, \quad \mu(\pi/2) = 1.$$

Then $(y\sin^2 x)' = 5\sin^2 x = \frac{5}{2}(1 - \cos 2x)$ (half-angle identity). Primitive: $\int \frac{5}{2}(1-\cos 2x)\,dx = \frac{5}{2}x - \frac{5}{4}\sin 2x$ (for the cosine term put $v = 2x$, $dv = 2\,dx$). Barrow from $\pi/2$ to x:

$$y\sin^2 x - 1\cdot 1 = \left[\frac{5}{2}s - \frac{5}{4}\sin 2s\right]_{\pi/2}^{x} = \frac{5}{2}x - \frac{5}{4}\sin 2x - \frac{5\pi}{4} + \frac{5}{4}\sin\pi.$$

Since $\sin\pi = 0$:

$$y(x) = \frac{1 + \frac{5}{2}x - \frac{5}{4}\sin 2x - \frac{5\pi}{4}}{\sin^2 x}, \qquad 0 < x < \pi.$$

Behavior at the ends of the interval: the numerator tends to $1 - \frac{5\pi}{4} \approx -2.927 \neq 0$ as $x \to 0^+$ and to $1 + \frac{5\pi}{4} \approx 4.927 \neq 0$ as $x \to \pi^-$, while $\sin^2 x \to 0$. So $y \to -\infty$ at 0^+ and $y \to +\infty$ at π^-: the maximal interval of existence is exactly $(0, \pi)$, where p is continuous.

Part (vii): $\dot{x} + 5x = t$

$p = 5, q = t, \mu = e^{5t}$, so $(xe^{5t})' = t\,e^{5t}$. Integrate by parts with $u = t, dv = e^{5t}dt, du = dt, v = e^{5t}/5$:

$$\int t\,e^{5t}\,dt = \frac{t\,e^{5t}}{5} - \int \frac{e^{5t}}{5}\,dt = \frac{t\,e^{5t}}{5} - \frac{e^{5t}}{25} + C.$$

Divide by e^{5t}:

$$x(t) = \frac{t}{5} - \frac{1}{25} + C\,e^{-5t}.$$

Part (viii): $\dot{x} + (a + 1/t)x = b,\ a > 0,\ x(1) = x_0$

On $t > 0$, with the three-step algorithm of the class notes [Notes, p. 3] ($p = a + 1/t, q = b, t_0 = 1$).

Step 1. $P(t) = \int_1^t \left(a + \frac{1}{s}\right) ds = [as + \ln s]_1^t = a(t-1) + \ln t$, so $I(t) = e^{a(t-1)+\ln t} = t\,e^{a(t-1)}$ and $I(1) = 1$.

Step 2. $(Ix)' = Iq = b\,t\,e^{a(t-1)}$; integrate from 1 to t:

$$\int_1^t \big(I(s)x(s)\big)'\,ds = b\int_1^t s\,e^{a(s-1)}\,ds.$$

Integration by parts ($u = s, du = ds, dv = e^{a(s-1)}ds, v = e^{a(s-1)}/a$):

$$\int_1^t s\,e^{a(s-1)}\,ds = \left[\frac{s\,e^{a(s-1)}}{a} - \frac{e^{a(s-1)}}{a^2}\right]_1^t = \frac{t\,e^{a(t-1)}}{a} - \frac{e^{a(t-1)}}{a^2} - \frac{1}{a} + \frac{1}{a^2}.$$

Step 3. Barrow's rule on the left gives $I(t)x(t) - I(1)x(1) = t\,e^{a(t-1)}x(t) - x_0$, hence

$$t\,e^{a(t-1)}x(t) = x_0 + b\left[\frac{t\,e^{a(t-1)}}{a} - \frac{e^{a(t-1)}}{a^2} - \frac{1}{a} + \frac{1}{a^2}\right].$$

Divide by $I(t) = t\,e^{a(t-1)}$ (the bracket's last two terms carry the factor $\frac{e^{-a(t-1)}}{t}$, like x_0):

$$x(t) = \frac{b}{a} - \frac{b}{a^2\,t} + \frac{e^{-a(t-1)}}{t}\left(x_0 - \frac{b}{a} + \frac{b}{a^2}\right).$$

Limit $t \to \infty$ with $a > 0$: $\frac{b}{a^2 t} \to 0$, and the last term is $\frac{K\,e^{-a(t-1)}}{t}$ with $K = x_0 - b/a + b/a^2$ constant; since $e^{-a(t-1)} \to 0$ and $1/t \to 0$ the product tends to 0. Therefore

$$\lim_{t\to\infty} x(t) = \frac{b}{a}.$$

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Substitution checks (each solution inserted into its equation):

- (i) $y = x^3/4 + C/x$: $y' = \frac{3x^2}{4} - \frac{C}{x^2}$, $\frac{y}{x} = \frac{x^2}{4} + \frac{C}{x^2}$; the sum is x^2.
- (ii) $x = 4 - 2e^{-t^2/2}$: $\dot{x} = 2t\,e^{-t^2/2}$, $tx = 4t - 2te^{-t^2/2}$; the sum is $4t$ and $x(0) = 2$. As $t \to \pm\infty$, $x \to 4$, the constant solution of $\dot{x} + tx = 4t$.
- (iii) $z = \frac{1}{2}\sin y \tan y + C\sec y$: sympy substitution of this expression gives $z' - z\tan y - \sin y = 0$.
- (iv) $y(0) = e^1[1 + 0] = e$; differentiating, $y' = -e^{-x}y + e^{e^{-x}}e^{-e^{-x}} = -e^{-x}y + 1$.
- (v) $x = 3\tanh t + C\,\mathrm{sech}\,t$: $\dot{x} = 3\,\mathrm{sech}^2 t - C\,\mathrm{sech}\,t\tanh t$ and $x\tanh t = 3\tanh^2 t + C\,\mathrm{sech}\,t\tanh t$; the sum is $3(\mathrm{sech}^2 t + \tanh^2 t) = 3$.
- (vi) at $x = \pi/2$: numerator $= 1 + \frac{5\pi}{4} - 0 - \frac{5\pi}{4} = 1$, denominator 1, so $y(\pi/2) = 1$; the ODE is satisfied (sympy residual 0).
- (vii) the particular solution $x_p = t/5 - 1/25$ satisfies $\dot{x}_p + 5x_p = \frac{1}{5} + t - \frac{1}{5} = t$, and Ce^{-5t} solves the homogeneous equation.
- (viii) $x(1) = \frac{b}{a} - \frac{b}{a^2} + x_0 - \frac{b}{a} + \frac{b}{a^2} = x_0$.

Dimensional remark for (viii), as a generic relaxation model with a a rate: $[a] = [t]^{-1}$ makes at dimensionless, and b/a has the dimensions of x, as required for the equilibrium of $\dot{x} = b - ax$. The term $1/t$ in the coefficient has the same dimensions as a.

RESULTS

Part	Integrating factor μ	Solution
(i)	x	$y = \frac{x^3}{4} + \frac{C}{x}$; finite at 0: $y = \frac{x^3}{4}$
(ii)	$e^{t^2/2}$	$x = 4 - 2e^{-t^2/2}$

Part	Integrating factor μ	Solution
(iii)	$\cos y$	$z = \frac{1}{2}\sin y \tan y + C\sec y$
(iv)	$e^{1-e^{-x}}$	$y = e^{e^{-x}}\left(1 + \int_0^x e^{-e^{-s}}\,ds\right)$
(v)	$\cosh t$	$x = 3\tanh t + C\,\mathrm{sech}\, t$
(vi)	$\sin^2 x$	$y = \frac{1+\frac{5}{2}x - \frac{5}{4}\sin 2x - \frac{5\pi}{4}}{\sin^2 x}$
(vii)	e^{5t}	$x = \frac{t}{5} - \frac{1}{25} + Ce^{-5t}$
(viii)	$t\,e^{at}$	$x = \frac{b}{a} - \frac{b}{a^2 t} + \frac{e^{-a(t-1)}}{t}\left(x_0 - \frac{b}{a} + \frac{b}{a^2}\right)$, limit b/a

EXERCISE 4

Exact Differential Equations

Original statement · ProblemsCh2.pdf, p. 2

Exactness test • Potential function

PROBLEM STATEMENT

Check that the following equations are exact and solve them.

(i) $(2xy - \sec^2 x) + (x^2 + 2y)\dfrac{dy}{dx} = 0,$

(ii) $(1 + e^x y + x e^x y) + (x e^x + 2)\dfrac{dy}{dx} = 0,$

(iii) $(x\cos y + \cos x)\dfrac{dy}{dx} + \sin y - y\sin x = 0,$

(iv) $e^x \sin y + y + (e^x \cos y + x + e^y)\dfrac{dy}{dx} = 0.$

Source: ProblemsCh2.pdf, Exercise 4 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Each equation is written in the form

$$M(x,y) + N(x,y)\,\frac{dy}{dx} = 0, \qquad \text{equivalently} \qquad M\,dx + N\,dy = 0.$$

The equation is exact if there is a function $F(x,y)$ of class C^2 with $F_x = M$ and $F_y = N$. Along any solution $y(x)$ one then has $\frac{d}{dx}F(x,y(x)) = F_x + F_y y' = M + Ny' = 0$, so $F(x,y(x)) = C$.

Hypotheses: M, N and their first partial derivatives continuous on a rectangle (or any simply connected open set) Ω. Then (Robinson, Section 10.1, condition (10.5)) the equation is exact on Ω if and only if

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}.$$

Part	M	N	Domain used
(i)	$2xy - \sec^2 x$	$x^2 + 2y$	$x \in (-\pi/2, \pi/2)$ (or any strip between zeros of $\cos x$), $y \in \mathbb{R}$
(ii)	$1 + e^x y + x e^x y$	$x e^x + 2$	$\mathbb{R}^2$

Part	M	N	Domain used
(iii)	$\sin y - y \sin x$	$x \cos y + \cos x$	$\mathbb{R}^2$
(iv)	$e^x \sin y + y$	$e^x \cos y + x + e^y$	$\mathbb{R}^2$

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: if the exactness criterion $M_y = N_x$ holds, a potential F exists and the solution curves are the level sets $F = C$, so no integrating factor or substitution is needed.

Construction of F, applied identically in each part:

- (a) Test: compute M_y and N_x and compare.
- (b) Integrate $F_x = M$ in x holding y fixed: $F = \int M\,dx + h(y)$, where the arbitrary function $h(y)$ plays the role of the constant of integration.
- (c) Differentiate in y and impose $F_y = N$: $h'(y) = N - \partial_y \int M\,dx$. The right side depends on y only because $M_y = N_x$.
- (d) Integrate h' and write the implicit solution $F(x, y) = C$.

METHOD OF THE CLASS NOTES [NOTES, P. 4]

The notes write the equation as $f(x, y) + g(x, y)\,\frac{dy}{dx} = 0$ (here $f = M, g = N$) and start from the answer: if $y = y(x)$ lies on a level set $F(x, y) = C$, the chain rule gives $F_x + F_y\,\frac{dy}{dx} = 0$, so an equation with $F_x = f, F_y = g$ is solved by $F = C$. Schwarz's theorem, $f_y = F_{yx} = F_{xy} = g_x$, gives the test, and the theorem of the notes reads: if f, g, f_y, g_x are continuous and $f_y = g_x$, the equation is exact.

The construction is steps (b) to (d) above, with the notes' $\varphi(y)$ in place of $h(y)$. Part (ii) is Ex 2 of the notes; there $\int xe^x\,dx = xe^x - e^x$ is done by parts ($u = x, dv = e^x\,dx$), which gives the same $F = x + xe^x y + 2y$ and $y = \frac{C - x}{xe^x + 2}$.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): $M = 2xy - \sec^2 x,\ N = x^2 + 2y$

(a) $M_y = 2x$ and $N_x = 2x$. They coincide, so the equation is exact.

(b) $F = \int (2xy - \sec^2 x)\,dx + h(y) = x^2 y - \tan x + h(y)$ (using $\frac{d}{dx} \tan x = \sec^2 x$).

(c) $F_y = x^2 + h'(y)$ must equal $N = x^2 + 2y$, so $h'(y) = 2y$.

(d) $h(y) = y^2$ (the constant is absorbed in C). Therefore

$$F(x, y) = x^2 y - \tan x + y^2 = C.$$

Solving the quadratic $y^2 + x^2 y - (\tan x + C) = 0$ for y gives, where the discriminant is nonnegative,

$$y(x) = \frac{-x^2 \pm \sqrt{x^4 + 4(\tan x + C)}}{2}.$$

Part (ii): $M = 1 + e^x y + x e^x y,\ N = x e^x + 2$

(a) $M_y = e^x + x e^x$ and $N_x = e^x + x e^x$ (product rule on $x e^x$). They coincide: exact.

(b) $F = \int [1 + y\,(e^x + x e^x)]\,dx + h(y)$. Since $\frac{d}{dx}(x e^x) = e^x + x e^x$, this integral is $x + y\,x e^x$. So $F = x + x e^x y + h(y)$.

(c) $F_y = x e^x + h'(y)$ must equal $x e^x + 2$, so $h'(y) = 2$.

(d) $h(y) = 2y$. Therefore

$$F(x, y) = x + x e^x y + 2y = C.$$

This is linear in y, so it can be solved explicitly:

$$y(x) = \frac{C - x}{x e^x + 2}.$$

The denominator is never zero: $g(x) = x e^x$ has $g'(x) = (1 + x)e^x = 0$ only at $x = -1$, where $g(-1) = -1/e \approx -0.368$, its global minimum, so $x e^x + 2 \geq 2 - 1/e > 0$. The solutions are therefore defined on all of $\mathbb{R}$.

Part (iii): $M = \sin y - y \sin x,\ N = x \cos y + \cos x$

(a) $M_y = \cos y - \sin x$ and $N_x = \cos y - \sin x$. Exact.

(b) $F = \int (\sin y - y \sin x)\,dx + h(y) = x \sin y + y \cos x + h(y)$ (since $\int \sin x\,dx = -\cos x$, so $-y \int \sin x\,dx = y \cos x$).

(c) $F_y = x \cos y + \cos x + h'(y)$ must equal $N = x \cos y + \cos x$, so $h'(y) = 0$ and h is constant.

(d) Absorbing the constant:

$$F(x, y) = x \sin y + y \cos x = C.$$

Part (iv): $M = e^x \sin y + y,\ N = e^x \cos y + x + e^y$

(a) $M_y = e^x \cos y + 1$ and $N_x = e^x \cos y + 1$. Exact.

(b) $F = \int (e^x \sin y + y)\, dx + h(y) = e^x \sin y + xy + h(y)$.

(c) $F_y = e^x \cos y + x + h'(y)$ must equal $e^x \cos y + x + e^y$, so $h'(y) = e^y$.

(d) $h(y) = e^y$. Therefore

$$F(x, y) = e^x \sin y + xy + e^y = C.$$

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check $F_x = M$ and $F_y = N$ for each potential:

- (i) $F = x^2 y - \tan x + y^2$: $F_x = 2xy - \sec^2 x = M$, $F_y = x^2 + 2y = N$.
- (ii) $F = x + xe^x y + 2y$: $F_x = 1 + (e^x + xe^x)y = M$, $F_y = xe^x + 2 = N$. Also, for $y = (C - x)/(xe^x + 2)$ with $C = 1$, $x = 0$: $y(0) = 1/2$; direct insertion gives $M(0, \frac{1}{2}) + N(0, \frac{1}{2})y'(0) = \frac{3}{2} + 2y'(0)$, and $y'(0) = -\frac{3}{4}$ from the quotient rule, so the sum is 0.
- (iii) $F = x \sin y + y \cos x$: $F_x = \sin y - y \sin x = M$, $F_y = x \cos y + \cos x = N$.
- (iv) $F = e^x \sin y + xy + e^y$: $F_x = e^x \sin y + y = M$, $F_y = e^x \cos y + x + e^y = N$.

Caveat on the level sets: $F(x, y) = C$ defines y as a function of x only where $F_y = N \neq 0$ (implicit function theorem). At points where $N = 0$ the level curve can have a vertical tangent and cannot be written as $y(x)$; for (ii) this never happens on the solution branch because $N = xe^x + 2 > 0$.

RESULTS

Part	Exactness	Solution
(i)	$M_y = N_x = 2x$	$x^2 y - \tan x + y^2 = C$
(ii)	$M_y = N_x = e^x(1 + x)$	$x + xe^x y + 2y = C$, i.e. $y = \frac{C - x}{xe^x + 2}$
(iii)	$M_y = N_x = \cos y - \sin x$	$x \sin y + y \cos x = C$
(iv)	$M_y = N_x = e^x \cos y + 1$	$e^x \sin y + xy + e^y = C$

EXERCISE 5

Integrating Factor for a Non-Exact Equation

Original statement · ProblemsCh2.pdf, p. 2

Non-exact • Integrating factor μ(x)

PROBLEM STATEMENT

Find an integrating factor depending only on x that makes the equation

$$3xy + y^2 + (x^2 + xy)\frac{dy}{dx} = 0$$

exact, and find its solution.

Source: ProblemsCh2.pdf, Exercise 5 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Write the equation as $M + N\,y' = 0$ with

$$M(x,y) = 3xy + y^2, \qquad N(x,y) = x^2 + xy = x(x+y).$$

M, N and their partial derivatives are polynomials, hence continuous on $\mathbb{R}^2$. Exactness test:

$$M_y = 3x + 2y, \qquad N_x = 2x + y.$$

$M_y - N_x = x + y$, which is not identically zero, so the equation is not exact. We look for $\mu = \mu(x)$ (nowhere zero on the domain considered) such that $\mu M + \mu N\,y' = 0$ is exact. We work on a region where $x \neq 0$ (the line $x = 0$ is excluded because $N(0,y) = 0$ there).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: multiplying the equation by a nonzero function does not change its solution set (on the region where the factor is nonzero) but can restore exactness. The requirement is $(\mu M)_y = (\mu N)_x$. If μ depends only on x, then $(\mu M)_y = \mu M_y$ and $(\mu N)_x = \mu' N + \mu N_x$, so the condition becomes

$$\mu M_y = \mu' N + \mu N_x \quad \Longleftrightarrow \quad \frac{\mu'}{\mu} = \frac{M_y - N_x}{N}.$$

This ordinary differential equation for $\mu(x)$ is solvable only if the right-hand side depends on x alone (otherwise no integrating factor of the form $\mu(x)$ exists). Then we integrate it, and continue with the standard exact-equation construction $F_x = \mu M$, $F_y = \mu N$.

CLASS NOTES AND COMPLEMENT [NOTES, P. 4]

The class notes solve exact equations only; the search for an integrating factor $\mu(x)$ that makes a non-exact equation exact is a complement (not in the class notes). Once μ is found, $\mu M + \mu N\, y' = 0$ is solved with the construction of the notes: test $f_y = g_x$ for $f = \mu M, g = \mu N$, integrate f in x and add $\varphi(y)$, and fix φ from $F_y = g$.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: find $\mu(x)$

Compute the quotient, factoring $N = x(x+y)$:

$$\frac{M_y - N_x}{N} = \frac{x+y}{x(x+y)} = \frac{1}{x} \qquad (x + y \neq 0).$$

This depends only on x, so μ exists. Solve $\mu'/\mu = 1/x$:

$$\int \frac{d\mu}{\mu} = \int \frac{dx}{x} \implies \ln|\mu| = \ln|x| + c \implies \mu(x) = x \quad \text{(taking the constant multiple equal to 1)}.$$

Step 2: check exactness of the new equation

Multiply by $\mu = x$:

$$\tilde{M} = xM = 3x^2 y + x y^2, \qquad \tilde{N} = xN = x^3 + x^2 y.$$

$$\tilde{M}_y = 3x^2 + 2xy, \qquad \tilde{N}_x = 3x^2 + 2xy.$$

They coincide, so the new equation is exact.

Step 3: potential function

Integrate $F_x = \tilde{M}$ in x at fixed y:

$$F = \int (3x^2y + xy^2)\,dx + h(y) = x^3y + \frac{x^2y^2}{2} + h(y).$$

Impose $F_y = \tilde{N}$:

$$F_y = x^3 + x^2y + h'(y) = x^3 + x^2y \implies h'(y) = 0.$$

So h is constant and the solution in implicit form is

$$x^3y + \frac{1}{2}x^2y^2 = C \qquad \Longleftrightarrow \qquad x^2y^2 + 2x^3y = K, \quad K = 2C.$$

Step 4: explicit form

Complete the square in y: $x^2y^2 + 2x^3y = x^2\left[(y+x)^2 - x^2\right]$. Then $x^2(y+x)^2 = K + x^4$, so

$$y(x) = -x \pm \frac{\sqrt{x^4 + K}}{|x|}, \qquad x \neq 0,\ x^4 + K \geq 0.$$

Remark: on the line $y = -x$ the coefficient $N = x(x+y)$ vanishes. Direct substitution of $y = -x$ gives $M = -3x^2 + x^2 = -2x^2$ and $N = 0$, so $M + Ny' = -2x^2 \neq 0$ and $y = -x$ is not a solution.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check the potential: $F = x^3y + \frac{1}{2}x^2y^2$ has $F_x = 3x^2y + xy^2 = x(3xy + y^2) = xM$ and $F_y = x^3 + x^2y = x(x^2 + xy) = xN$, as required.

Check along a solution curve: implicit differentiation of $x^3y + \frac{1}{2}x^2y^2 = C$ gives $(3x^2y + xy^2) + (x^3 + x^2y)y' = 0$. Dividing by $x \neq 0$ gives back $(3xy + y^2) + (x^2 + xy)y' = 0$, the original equation. The division by x is why the line $x = 0$ must be excluded from the region of validity.

Numeric spot check of the explicit branch: with $K = 3$ and $x = 1$, $y = -1 + \sqrt{1+3} = 1$; then $F = 1 + \frac{1}{2} = \frac{3}{2} = K/2$, consistent. The slope from the ODE is $y' = -\frac{3xy + y^2}{x^2 + xy} = -\frac{4}{2} = -2$, and the derivative of $y = -x + \sqrt{x^4 + 3}/x$ is $-1 + \frac{x^4 - 3}{x^2\sqrt{x^4+3}}$, which equals $-1 + \frac{-2}{1 \cdot 2} = -2$ at $x = 1$.

RESULT

Integrating factor $\mu(x) = x$ (up to a constant factor). Solution: $x^3 y + \frac{1}{2} x^2 y^2 = C$ on regions with $x \neq 0$.

EXERCISE 6

Exactness of Separated Differential Forms

Original statement · ProblemsCh2.pdf, p. 2

Exact • Separated form f(x) + g(y)y'

PROBLEM STATEMENT

Show that any equation that can be written in the form

$$f(x) + g(y)\frac{dy}{dx} = 0$$

is exact, and find its solution in terms of integrals of f and g. Hence find the solutions of

(i) $V'(x) + 2y\dfrac{dy}{dx} = 0,$
(ii) $\left[\dfrac{1}{y} - a\right]\dfrac{dy}{dx} + \dfrac{2}{x} - b = 0,$ for $x, y > 0.$

Source: ProblemsCh2.pdf, Exercise 6 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

General form: $M(x,y) + N(x,y)\,y' = 0$ with $M = f(x)$ depending on x only and $N = g(y)$ depending on y only. Assume f continuous on an interval I_x and g continuous on an interval I_y, so that M and N are continuous on the rectangle $\Omega = I_x \times I_y$ (a simply connected set).

- (i) $f(x) = V'(x)$, with $V \in C^1$ so that V' is continuous; $g(y) = 2y$. Rectangle: I_x any interval, $I_y = \mathbb{R}$.
- (ii) $f(x) = \frac{2}{x} - b$, $g(y) = \frac{1}{y} - a$. Continuous on $x > 0$ and $y > 0$ (the stated domain), so $\Omega = (0,\infty) \times (0,\infty)$.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: the exactness criterion $M_y = N_x$ (Robinson, Section 10.1) is the hypothesis for a potential function F with $F_x = M$, $F_y = N$ to exist on Ω. Here both partial derivatives are trivial, so the criterion can be checked at once, and the potential can be constructed by integrating each variable separately.

Construction: integrate $F_x = f(x)$ in x, then fix the y-dependent integration function $h(y)$ from $F_y = g(y)$. Level sets of F are the solutions.

METHOD OF THE CLASS NOTES [NOTES, PP. 3-4]

In the notation of the notes the equation is $f + g\,\frac{dy}{dx} = 0$ with $f = f(x)$ and $g = g(y)$, so $\frac{\partial f}{\partial y} = 0 = \frac{\partial g}{\partial x}$ and the theorem of the notes (with f, g continuous) gives exactness. The construction of the notes then reads $F = \int f(x)\,dx + \varphi(y)$, $F_y = \varphi'(y) = g(y)$, so $F = \int f(x)\,dx + \int g(y)\,dy = C$: the separable formula of p. 3 written as a level set.

PHASE 3: STEP-BY-STEP DERIVATION

General case

$M = f(x)$ does not depend on y, and $N = g(y)$ does not depend on x:

$$\frac{\partial M}{\partial y} = 0 = \frac{\partial N}{\partial x}.$$

The exactness condition holds, so the equation is exact. Construct F: from $F_x = f(x)$,

$$F(x, y) = \int_{x_0}^{x} f(s)\,ds + h(y).$$

Differentiate in y: $F_y = h'(y)$, and this must equal $N = g(y)$. So $h'(y) = g(y)$ and

$$h(y) = \int_{y_0}^{y} g(r)\,dr.$$

Hence the solutions are the level curves

$$\boxed{\int_{x_0}^{x} f(s)\,ds + \int_{y_0}^{y} g(r)\,dr = C}, \qquad C \in \mathbb{R}.$$

If a solution passes through (x_0, y_0), then $C = 0$ with these lower limits (Barrow's rule fixes the constant). The same result follows from separation of variables: $g(y)\,dy = -f(x)\,dx$, then integrate both sides.

Part (i): $V'(x) + 2y\,y' = 0$

Here $f = V'$ and $g = 2y$. The integrals are $\int V'(x)\,dx = V(x)$ and $\int 2y\,dy = y^2$. Therefore

$$V(x) + y^2 = C.$$

Equivalent explicit form: $y = \pm\sqrt{C - V(x)}$, defined where $V(x) \le C$. The equation says $\frac{d}{dx}\left[V(x) + y^2\right] = V' + 2yy' = 0$, i.e. $V + y^2$ is conserved along the solution.

Part (ii): $\left[\frac{1}{y} - a\right] y' + \frac{2}{x} - b = 0, \quad x, y > 0$

Here $f(x) = \frac{2}{x} - b$ and $g(y) = \frac{1}{y} - a$. Integrate each, using $x > 0$ and $y > 0$ so no absolute values are needed:

$$\int \left(\frac{2}{x} - b\right) dx = 2\ln x - bx, \qquad \int \left(\frac{1}{y} - a\right) dy = \ln y - ay.$$

The solutions are therefore

$$2\ln x - bx + \ln y - ay = C.$$

Combine the logarithms and exponentiate: $\ln(x^2 y) = C + bx + ay$, so with $K = e^C > 0$,

$$x^2\, y\, e^{-bx - ay} = K.$$

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check by implicit differentiation:

- General case: $\frac{d}{dx}\left[\int_{x_0}^{x} f\, ds + \int_{y_0}^{y(x)} g\, dr\right] = f(x) + g(y)\, y'$, using the Fundamental Theorem of Calculus and the chain rule; this vanishes by the ODE.
- (i) $\frac{d}{dx}\left[V(x) + y^2\right] = V'(x) + 2y\, y' = 0$.
- (ii) $\frac{d}{dx}\left[2\ln x - bx + \ln y - ay\right] = \frac{2}{x} - b + \left(\frac{1}{y} - a\right) y' = 0$, the original equation.

Dimensional remark for (ii): $\ln x$ and $\ln y$ require dimensionless arguments, so x and y are understood as measured in fixed reference units; then bx and ay are dimensionless, which means $[b] = [x]^{-1}$ and $[a] = [y]^{-1}$, consistent with the terms $\frac{2}{x} - b$ and $\frac{1}{y} - a$ being summed.

Domain: in (ii) the level set $x^2 y e^{-bx-ay} = K$ in the open quadrant is a union of curves; y is a function of x locally wherever $F_y = \frac{1}{y} - a \neq 0$, i.e. away from $y = 1/a$ when $a > 0$.

RESULTS

General form: $\int f\,dx + \int g\,dy = C$. (i) $V(x) + y^2 = C$. (ii) $2\ln x - bx + \ln y - ay = C$, i.e. $x^2 y\, e^{-bx-ay} = K > 0$.

EXERCISE 7

Nonlinear Change of Variables

Original statement · ProblemsCh2.pdf, p. 2

Substitution z = y² • Linear reduction

PROBLEM STATEMENT

> We consider the following initial value problem:
>
> $$y'(x) = y(x) + \frac{x}{y(x)}, \qquad y(0) = 1.$$
>
> Make the change of variable $z(x) = y(x)^2$ and find the solution.

Source: ProblemsCh2.pdf, Exercise 7 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Right-hand side: $f(x, y) = y + x/y$. It is continuous for $y \neq 0$, and $\partial f/\partial y = 1 - x/y^2$ is continuous for $y \neq 0$. The initial point $(0, 1)$ lies in the region $y > 0$, so Theorem 6.2 gives a unique local solution. The line $y = 0$ is excluded from the domain, so the solution cannot cross it: while it exists it keeps the sign of $y(0) = 1$, that is $y(x) > 0$.

The equation is non-linear in y because of the term $x\,y^{-1}$. It has Bernoulli form $y' - y = x\,y^{-1}$ with exponent $\alpha = -1$ (see Problem 2.8).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this substitution: multiplying the equation by y gives $y\,y' = y^2 + x$, and the left side is $\frac{1}{2}\frac{d}{dx}(y^2)$. In terms of $z = y^2$ the right side $y^2 + x = z + x$ is linear, so the problem becomes a linear first-order equation, solvable with an integrating factor.

Change of variables written explicitly: $z = y^2$, $z' = 2y\,y'$ (chain rule), $z(0) = y(0)^2 = 1$. Back-substitution: since $y > 0$, $y = +\sqrt{z}$.

CLASS NOTES AND COMPLEMENT [NOTES, P. 3]

The substitution $z = y^2$ is a complement (not in the class notes). The linear equation it produces, $z' - 2z = 2x$, $z(0) = 1$, is solved in Step 2 below with the three-step algorithm of the notes (μ there is the notes' I): Step 1, $P(x) = \int_0^x (-2)\,ds = -2x$, $I(x) = e^{-2x}$, $I(0) = 1$; Step 2, $\int_0^x (Iz)'\,ds = \int_0^x 2s\,e^{-2s}\,ds$; Step 3, $z(x) = \frac{z(0)}{I(x)} + \frac{1}{I(x)}\int_0^x I(s)\,2s\,ds$.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: transform the equation

Multiply $y' = y + x/y$ by $2y$ (allowed since $y \neq 0$):

$$2y\,y' = 2y^2 + 2x.$$

With $z = y^2$ and $z' = 2yy'$:

$$z' = 2z + 2x, \qquad z(0) = 1, \qquad \text{i.e.} \quad z' - 2z = 2x.$$

Step 2: integrating factor

$p(x) = -2$, so $\int p\,dx = -2x$ and $\mu(x) = e^{-2x}$, $\mu(0) = 1$. Then $(z\,e^{-2x})' = 2x\,e^{-2x}$. Integrate from 0 to x (Barrow):

$$z(x)e^{-2x} - z(0)\cdot 1 = \int_0^x 2s\,e^{-2s}\,ds.$$

Integration by parts with $u = 2s$, $dv = e^{-2s}ds$, $du = 2\,ds$, $v = -\frac{1}{2}e^{-2s}$:

$$\int 2s\,e^{-2s}\,ds = -s\,e^{-2s} + \int e^{-2s}\,ds = -s\,e^{-2s} - \frac{1}{2}e^{-2s}.$$

Evaluate between the limits 0 and x:

$$\left[-s\,e^{-2s} - \tfrac{1}{2}e^{-2s}\right]_0^x = -x\,e^{-2x} - \tfrac{1}{2}e^{-2x} + \tfrac{1}{2}.$$

Hence $z\,e^{-2x} = 1 - x\,e^{-2x} - \frac{1}{2}e^{-2x} + \frac{1}{2} = \frac{3}{2} - x\,e^{-2x} - \frac{1}{2}e^{-2x}$. Multiply by e^{2x}:

$$z(x) = \frac{3}{2}e^{2x} - x - \frac{1}{2}.$$

Step 3: return to y and find the domain

Since $y > 0$ and $y^2 = z$, we need $z > 0$ and take the positive root:

$$y(x) = \sqrt{\frac{3}{2}e^{2x} - x - \frac{1}{2}}.$$

Domain: is $z(x) > 0$ for all x? Minimize z: $z'(x) = 3e^{2x} - 1 = 0$ at $e^{2x} = 1/3$, i.e. $x_m = -\frac{1}{2}\ln 3 \approx -0.549$. Since $z'' = 6e^{2x} > 0$ this is the global minimum, and

$$z(x_m) = \frac{3}{2}\cdot\frac{1}{3} + \frac{1}{2}\ln 3 - \frac{1}{2} = \frac{1}{2}\ln 3 \approx 0.549 > 0.$$

So $z(x) \geq \frac{1}{2}\ln 3 > 0$ for every real x: the solution never reaches $y = 0$ and is defined on all of $\mathbb{R}$ (the maximal interval of existence is $\mathbb{R}$).

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Substitute back into the original equation. Differentiate $y = \sqrt{z}$: $y' = \dfrac{z'}{2y} = \dfrac{3e^{2x} - 1}{2y}$. The right-hand side is

$$y + \frac{x}{y} = \frac{y^2 + x}{y} = \frac{z + x}{y} = \frac{\frac{3}{2}e^{2x} - \frac{1}{2}}{y} = \frac{3e^{2x} - 1}{2y}.$$

They agree. Initial condition: $z(0) = \frac{3}{2} - 0 - \frac{1}{2} = 1$, so $y(0) = 1$.

- As $x \to +\infty$: $y \sim \sqrt{3/2}\, e^{x}$, exponential growth, consistent with $y' \approx y$ for large y.
- As $x \to -\infty$: $z \sim -x$, so $y \sim \sqrt{|x|} \to \infty$: the term x/y is negative for $x < 0$, while $z \approx -x$ grows.

Numeric check (fourth-order Runge-Kutta against the formula): the maximum absolute difference at $x = -2, -1, 1, 2$ is about 4.5×10^{-13} (fourth-order Runge-Kutta, 40000 steps per interval).

RESULT

$$y(x) = \sqrt{\tfrac{3}{2}e^{2x} - x - \tfrac{1}{2}}, \qquad x \in \mathbb{R}.$$

EXERCISE 8

General Bernoulli Equation Reduction

Original statement · ProblemsCh2.pdf, p. 2

Bernoulli • Substitution z = y^(1-a)

PROBLEM STATEMENT

Show that the Bernoulli equation

$$y'(x) = a(x)\,y(x) + b(x)\,y(x)^{\alpha},$$

where $\alpha \neq 0, 1$, can be transformed into a linear ODE with the change of variable $z(x) = y(x)^{1-\alpha}$ and find the solution. What happens if $\alpha = 0$ or $\alpha = 1$?

Source: ProblemsCh2.pdf, Exercise 8 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Assume a and b are continuous on an interval I and $\alpha \in \mathbb{R}$, $\alpha \neq 0, 1$. The power y^{α} and $y^{1-\alpha}$ must make sense, so we work with $y > 0$ (for rational α with odd denominators other branches can be treated analogously). On $y > 0$, $f(x, y) = ay + by^{\alpha}$ and $\partial f/\partial y = a + \alpha b\, y^{\alpha-1}$ are continuous, so IVPs with $y(x_0) = y_0 > 0$ have unique local solutions.

The equation is non-linear because y^{α} with $\alpha \neq 0, 1$ is not affine in y. When $\alpha > 0$, $y \equiv 0$ is also a solution; it is excluded by the assumption $y > 0$ and has to be added by hand (for $0 < \alpha < 1$ uniqueness at $y = 0$ may fail, as in Problem 2.14).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this substitution: dividing by y^α gives $y^{-\alpha}y' = a\,y^{1-\alpha} + b$, where the left side is, up to a constant factor, the derivative of $y^{1-\alpha}$ and the right side contains $y^{1-\alpha}$ linearly. So the natural unknown is $z = y^{1-\alpha}$.

Change of variables written explicitly: $z = y^{1-\alpha},\ z' = (1-\alpha)\,y^{-\alpha}\,y'$ (chain rule), $y = z^{1/(1-\alpha)}$.

CLASS NOTES AND COMPLEMENT [NOTES, P. 3]

The Bernoulli substitution $z = y^{1-\alpha}$ is a complement (not in the class notes). The linear equation for z is then solved in Step 2 below with the three-step algorithm of the notes (μ there is the notes' I): Step 1 takes $P(x) = \int_{x_0}^{x} p(s)\,ds = -(1-\alpha)A(x)$, so $I(x_0) = 1$; Step 2 integrates $(Iz)' = Iq$ from x_0 to x; Step 3 gives $z = \frac{z_0}{I} + \frac{1}{I}\int_{x_0}^{x} Iq\,ds$, which is the formula of Step 2.

PHASE 3: STEP-BY-STEP DERIVATION

Step 1: derive the linear equation

Differentiate $z = y^{1-\alpha}$ with the chain rule and substitute the equation for y':

$$z' = (1-\alpha)\,y^{-\alpha}\,y' = (1-\alpha)\,y^{-\alpha}\,(a\,y + b\,y^{\alpha}) = (1-\alpha)\left(a\,y^{1-\alpha} + b\right).$$

Since $y^{1-\alpha} = z$:

$$z' - (1-\alpha)\,a(x)\,z = (1-\alpha)\,b(x).$$

This is linear first order in z, with $p(x) = -(1-\alpha)a(x)$ and $q(x) = (1-\alpha)b(x)$, both continuous.

Step 2: solve it with an integrating factor

Fix $x_0 \in I$ and define $A(x) = \int_{x_0}^{x} a(s)\,ds$. Then $\int p\,dx = -(1-\alpha)A(x)$ and

$$\mu(x) = e^{-(1-\alpha)A(x)}, \qquad \mu(x_0) = 1, \qquad (\mu z)' = (1-\alpha)\,b(x)\,\mu(x).$$

Integrate from x_0 to x (Barrow's rule), with $z_0 = z(x_0) = y_0^{1-\alpha}$:

$$\mu(x)z(x) - z_0 = (1-\alpha)\int_{x_0}^{x} b(s)\, e^{-(1-\alpha)A(s)}\, ds.$$

Multiply by $1/\mu(x) = e^{(1-\alpha)A(x)}$:

$$z(x) = e^{(1-\alpha)A(x)}\left[z_0 + (1-\alpha)\int_{x_0}^{x} b(s)\, e^{-(1-\alpha)A(s)}\, ds\right].$$

Step 3: return to y

Since $y = z^{1/(1-\alpha)}$, wherever the bracket is positive (so that $y > 0$):

$$y(x) = \left\{e^{(1-\alpha)A(x)}\left[y_0^{1-\alpha} + (1-\alpha)\int_{x_0}^{x} b(s)\, e^{-(1-\alpha)A(s)}\, ds\right]\right\}^{\frac{1}{1-\alpha}}.$$

The solution exists as long as the expression in braces stays positive; if it reaches 0 at a finite point the solution ceases to exist there (the substitution is valid only for $y > 0$).

Step 4: the excluded cases $\alpha = 0$ and $\alpha = 1$

- $\alpha = 1$: the equation is $y' = (a+b)\,y$, already linear and homogeneous. The substitution would give $z = y^0 = 1$, a constant that carries no information (the formula above has the factor $1-\alpha = 0$, so it degenerates to $z' = 0$). The solution is obtained directly: $y(x) = y_0 \exp\left(\int_{x_0}^{x}(a+b)\, ds\right)$.
- $\alpha = 0$: the equation is $y' = a\,y + b$, already linear (inhomogeneous). The substitution gives $z = y^1 = y$, the identity, and the formula above with $1-\alpha = 1$ reproduces the variation-of-constants solution $y = e^{A}\left[y_0 + \int_{x_0}^{x} b\, e^{-A}\, ds\right]$. No transformation is needed.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check the general solution by differentiation. Put $B(x) = z_0 + (1-\alpha)\int_{x_0}^{x} b\, e^{-(1-\alpha)A}\, ds$, so $z = e^{(1-\alpha)A}B$ and

$$z' = (1-\alpha)a\, e^{(1-\alpha)A}B + e^{(1-\alpha)A}(1-\alpha)\, b\, e^{-(1-\alpha)A} = (1-\alpha)a\, z + (1-\alpha)b.$$

This is the linear equation of Step 1, and $z(x_0) = e^0 z_0 = z_0$. Undoing the change of variables is reversible for $y > 0$, so $y = z^{1/(1-\alpha)}$ solves the Bernoulli equation.

Consistency with Problem 2.7: for $a=1, b=x, \alpha=-1$ the formula gives $1-\alpha=2$, $A=x$, $z'-2z=2x$, which is the equation obtained there for $z=y^2$.

Check with a constant-coefficient case (a, b constants, $\alpha=2$, so $1-\alpha=-1$): $z'+az=-b$, giving $z=-\frac{b}{a}+\left(z_0+\frac{b}{a}\right)e^{-a(x-x_0)}$ and $y=1/z$, which is the logistic-type solution of $y'=ay+by^2$. A sympy substitution of this closed form into $y'=ay+by^2$ (with $a, b>0$ symbols and $y_0=1$) gives residual 0.

RESULT

$z=y^{1-\alpha}$ turns the Bernoulli equation into $z'-(1-\alpha)az=(1-\alpha)b$, with solution $z=e^{(1-\alpha)A}\left[z_0+(1-\alpha)\int b\,e^{-(1-\alpha)A}\right]$ and $y=z^{1/(1-\alpha)}$. For $\alpha=1$ and $\alpha=0$ the equation is already linear and the substitution is trivial or degenerate.

EXERCISE 9

Uniqueness and Lipschitz Analysis

Original statement · ProblemsCh2.pdf, p. 2

Existence-uniqueness • Non-Lipschitz points

PROBLEM STATEMENT

Which of the following differential equations have unique solutions (at least on some small time interval) for any non-negative initial condition $x(0) \geq 0$?

(i) $\dot{x} = x(1 - x^2)$,
(ii) $\dot{x} = x^3$,
(iii) $\dot{x} = x^{1/3}$,
(iv) $\dot{x} = x^{1/2}(1 + x)^2$,
(v) $\dot{x} = (1 + x)^{3/2}$.

Source: ProblemsCh2.pdf, Exercise 9 (page 2).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Each equation is autonomous: $\dot{x} = f(x)$ with initial condition $x(0) = x_0 \geq 0$.

Uniqueness criterion used (Robinson, Theorem 6.2): if f and $\partial f/\partial x$ are continuous in an open neighborhood of (t_0, x_0), then the IVP has a unique solution on some open interval containing t_0. This is a sufficient condition, not a necessary one: a more general sufficient condition is that f be locally Lipschitz in x, $|f(x) - f(y)| \leq L|x - y|$. When the condition fails at a point, uniqueness there must be examined directly.

Part	$f(x)$	$f'(x)$	**Where f' is continuous**
(i)	$x - x^3$	$1 - 3x^2$	all x
(ii)	x^3	$3x^2$	all x
(iii)	$x^{1/3}$	$\frac{1}{3}x^{-2/3}$	$x \neq 0$
(iv)	$x^{1/2}(1+x)^2$	$\frac{1}{2}x^{-1/2}(1+x)^2 + 2x^{1/2}(1+x)$	$x > 0$
(v)	$(1+x)^{3/2}$	$\frac{3}{2}(1+x)^{1/2}$	$x > -1$ (so all $x \geq 0$)

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: for each f, check where f and f' are continuous. On those sets Theorem 6.2 applies. At any point $x_0 \geq 0$ where it does not apply (here only $x_0 = 0$ in (iii) and (iv), where f' is unbounded), decide uniqueness by constructing two solutions (separation of variables) or by showing that they cannot exist.

For a decisive non-uniqueness test at $x_0 = 0$ with $f(0) = 0$ and $f > 0$ for $x > 0$: the constant $x \equiv 0$ is always a solution, and by separation of variables, $t = \int_0^x ds/f(s)$ defines a second solution $x(t) > 0$ for $t > 0$ whenever the integral $\int_0^\varepsilon ds/f(s)$ is finite (the time needed to leave 0 is then finite).

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The class notes state Picard's theorem: if f and $\partial f/\partial y$ are continuous near (x_0, y_0), there is $h > 0$ such that $y' = f(x, y), y(x_0) = y_0$ has a unique solution for $|x - x_0| < h$. Here the unknown is $x(t)$ and $f = f(x)$, so the test is the continuity of f and $f'(x)$ near the initial value: this is the check made below (Robinson's Theorem 6.2 is the same statement).

The theorem is a sufficient condition only. Where f' is not continuous it says nothing, and uniqueness must be decided by hand, as in (iii) and (iv) at $x_0 = 0$.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): $\dot{x} = x(1 - x^2)$

$f(x) = x - x^3$ and $f'(x) = 1 - 3x^2$ are polynomials, continuous everywhere. Theorem 6.2 applies at every $x_0 \geq 0$. Answer: unique solution for all $x_0 \geq 0$.

Part (ii): $\dot{x} = x^3$

$f = x^3$, $f' = 3x^2$, continuous everywhere. Answer: unique solution for all $x_0 \geq 0$ (locally in time). Remark: the solution blows up in finite time. Separating, $\int_{x_0}^x s^{-3}\, ds = t$ gives $-\frac{1}{2x^2} + \frac{1}{2x_0^2} = t$, so $x(t) = \left(x_0^{-2} - 2t\right)^{-1/2}$ for $x_0 > 0$, which exists only for $t < T^* = \frac{1}{2x_0^2}$. This is why the problem says 'at least on some small time interval'.

Part (iii): $\dot{x} = x^{1/3}$

$f = x^{1/3}$ is continuous, but $f'(x) = \frac{1}{3}x^{-2/3} \to \infty$ as $x \to 0^+$. For $x_0 > 0$, f and f' are continuous near x_0, so the solution is unique. For $x_0 = 0$ the theorem does not apply, and uniqueness in fact fails: the constant $x \equiv 0$ is a solution, and separating variables with $x \geq 0$,

$$\int_0^x s^{-1/3}\, ds = \left[\tfrac{3}{2}s^{2/3}\right]_0^x = \tfrac{3}{2}x^{2/3} = t \quad \Longrightarrow \quad x(t) = \left(\tfrac{2t}{3}\right)^{3/2}, \quad t \geq 0,$$

is a second solution with $x(0)=0$. Check: $\dot{x}=\frac{3}{2}\cdot\frac{2}{3}\left(\frac{2t}{3}\right)^{1/2}=\left(\frac{2t}{3}\right)^{1/2}$ and $x^{1/3}=\left(\frac{2t}{3}\right)^{1/2}$. So there are at least two solutions from $x_0=0$ (indeed infinitely many, obtained by staying at 0 until any time $c\geq 0$ and then following $\left(\frac{2(t-c)}{3}\right)^{3/2}$). Answer: unique for $x_0>0$, not unique for $x_0=0$.

Part (iv): $\dot{x}=x^{1/2}(1+x)^2$

$f(x)=\sqrt{x}\,(1+x)^2$ is continuous on $x\geq 0$ and $f'(x)=\frac{(1+x)^2}{2\sqrt{x}}+2\sqrt{x}\,(1+x)\to\infty$ as $x\to 0^+$. For $x_0>0$ Theorem 6.2 applies and the solution is unique. At $x_0=0$ the theorem does not apply. We show non-uniqueness: besides $x\equiv 0$, separate variables for $x>0$ and substitute $s=r^2$, $ds=2r\,dr$ (so $\sqrt{s}=r$):

$$t=\int_0^x\frac{ds}{\sqrt{s}\,(1+s)^2}=\int_0^{\sqrt{x}}\frac{2\,dr}{(1+r^2)^2}=\left[\frac{r}{1+r^2}+\arctan r\right]_0^{\sqrt{x}}=\frac{\sqrt{x}}{1+x}+\arctan\sqrt{x}.$$

The primitive is checked by differentiation: $\frac{d}{dr}\left[\frac{r}{1+r^2}+\arctan r\right]=\frac{1-r^2}{(1+r^2)^2}+\frac{1}{1+r^2}=\frac{2}{(1+r^2)^2}$. The function $T(x)=\frac{\sqrt{x}}{1+x}+\arctan\sqrt{x}$ is continuous and strictly increasing on $x\geq 0$ (its derivative is $1/(\sqrt{x}(1+x)^2)>0$), with $T(0)=0$ and $T(x)\to\pi/2$ as $x\to\infty$. Its inverse $x(t)=T^{-1}(t)$, $0\leq t<\pi/2$, is a solution with $x(0)=0$ and $x(t)>0$ for $t>0$. So at $x_0=0$ there are at least two solutions. Answer: unique for $x_0>0$, not unique for $x_0=0$.

Part (v): $\dot{x}=(1+x)^{3/2}$

$f=(1+x)^{3/2}$ and $f'=\frac{3}{2}(1+x)^{1/2}$ are continuous for $x>-1$, in particular for every $x_0\geq 0$. Answer: unique solution for all $x_0\geq 0$. The solution exists only up to a finite time: separating,

$$\int_{x_0}^{x}(1+s)^{-3/2}\,ds=\left[-2(1+s)^{-1/2}\right]_{x_0}^{x}=-\frac{2}{\sqrt{1+x}}+\frac{2}{\sqrt{1+x_0}}=t,$$

so $x(t)=\left(\frac{1}{\sqrt{1+x_0}}-\frac{t}{2}\right)^{-2}-1$, which blows up at $T^*=\frac{2}{\sqrt{1+x_0}}$ (for $x_0=0$, $T^*=2$).

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Summary of the criterion check and numerical cross-checks: for (iii), with $x(t)=(2t/3)^{3/2}$ the residual $\dot{x}-x^{1/3}$ evaluated at $t=0.7, 1.3, 5$ is of order 10^{-17} (rounding); for (iv), sympy gives $T'(x)\,f(x)=1$ identically; for (ii) and (v) the closed forms give residual 0 symbolically.

Non-uniqueness and the Lipschitz condition are consistent: in (iii) and (iv), $\frac{|f(x)-f(0)|}{x}=x^{-2/3}$ and $\sim x^{-1/2}$ respectively, which are unbounded as $x\to 0^+$, so no Lipschitz constant exists near 0. (For (iii) and (iv) the integral $\int_0 ds/f(s)$ converges, which is exactly what allows the solution to leave 0 in finite time.) By contrast, in (i), (ii), (v) f is Lipschitz on every bounded interval, and uniqueness holds.

ANSWER

Part	Unique for every $x_0 \geq 0$?	Comment
(i)	Yes	$f \in C^1$; forward in time the solutions stay bounded (they tend to the equilibria 0 or 1). Backward in time, for $x_0 > 1$ the solution escapes to $+\infty$ in finite negative time, because $\int^{\infty} dx/(x - x^3)$ converges
(ii)	Yes (locally in time)	blow-up at $T^* = 1/(2x_0^2)$ for $x_0 > 0$
(iii)	No	fails at $x_0 = 0$: $x \equiv 0$ and $x = (2t/3)^{3/2}$
(iv)	No	fails at $x_0 = 0$: $x \equiv 0$ and $x = T^{-1}(t)$
(v)	Yes (locally in time)	blow-up at $T^* = 2/\sqrt{1 + x_0}$

EXERCISE 10

Uniqueness via an Integrating Transformation

Original statement · ProblemsCh2.pdf, p. 3

Linear homogeneous • Uniqueness without Picard

PROBLEM STATEMENT

Without using Picard's theorem, show that the IVP

$$\frac{dy}{dt} + p(t)\, y = 0, \qquad y(t_0) = y_0$$

with $p(t)$ a continuous function has a unique solution, and calculate it. What solution do we obtain if $y_0 = 0$? Hint: consider the function $z(t) = y(t)\, e^{\int_{t_0}^{t} p(s)\, ds}$, where $y(t)$ solves the IVP, show that $z(t)$ is constant and deduce $y(t)$.

Source: ProblemsCh2.pdf, Exercise 10 (page 3).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Data: p continuous on an open interval $I \ni t_0$; $y_0 \in \mathbb{R}$ arbitrary. A solution is a differentiable function $y : J \to \mathbb{R}$ on an interval $J \subseteq I$ containing t_0 with $y' = -p(t)\, y$ and $y(t_0) = y_0$. Linearity of the equation means that no non-linear terms can cause finite-time blow-up, so we may expect solutions on all of I.

We must prove two statements: (a) any solution is given by one explicit formula (this gives uniqueness), (b) that formula really is a solution (existence). Picard's theorem is not to be used; only the Fundamental Theorem of Calculus and the fact that a function with zero derivative on an interval is constant (Mean Value Theorem) are allowed.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method (the hint): the function $P(t) = \int_{t_0}^{t} p(s)\, ds$ is differentiable with $P' = p$ because p is continuous (Fundamental Theorem of Calculus). The function $e^{P(t)}$ is exactly an integrating factor μ for $y' + py = 0$. For any solution y, the product $z = y\, e^{P}$ should have zero derivative, which forces z to be constant.

Change of unknown, written explicitly: $z(t) = y(t)\, e^{P(t)}$, $P(t) = \int_{t_0}^{t} p(s)\, ds$, $P(t_0) = 0$, hence $z(t_0) = y_0$.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The hint is Step 1 of the three-step algorithm of the notes with $q = 0$: $P(t) = \int_{t_0}^{t} p(s)\,ds$, $P(t_0) = 0$, $I(t_0) = 1$, and $z = I\,y$. Step 2 with $q = 0$ is $(Iy)' = 0$, and Step 3 gives $y(t) = \frac{y_0}{I(t)} = y_0\,e^{-P(t)}$. What the notes take for granted (that every solution has this form) is what this problem proves: the zero derivative forces z to be constant by the Mean Value Theorem, with no appeal to Picard.

PHASE 3: STEP-BY-STEP DERIVATION

Uniqueness

Let y be any solution. Differentiate $z = y\,e^{P}$ with the product rule and the chain rule ($\frac{d}{dt}e^{P} = P'\,e^{P} = p\,e^{P}$):

$$z'(t) = y'(t)\,e^{P(t)} + y(t)\,p(t)\,e^{P(t)} = e^{P(t)}\,[y'(t) + p(t)\,y(t)] = e^{P(t)} \cdot 0 = 0.$$

A function whose derivative vanishes on an interval is constant (Mean Value Theorem), so $z(t) = z(t_0)$ for all $t \in J$. Evaluate at t_0: $z(t_0) = y(t_0)\,e^{0} = y_0$. Hence $y(t)\,e^{P(t)} = y_0$, and since $e^{P(t)} > 0$,

$$y(t) = y_0\,e^{-P(t)} = y_0 \exp\left(-\int_{t_0}^{t} p(s)\,ds\right).$$

Every solution coincides with this expression, so there is at most one solution.

Existence

Define $y(t) = y_0\,e^{-P(t)}$ on I. It is differentiable, with $y' = y_0\,e^{-P} \cdot (-P') = -p(t)\,y(t)$, i.e. $y' + py = 0$, and $y(t_0) = y_0 e^{0} = y_0$. So the IVP has this solution on all of I.

The case $y_0 = 0$

The formula gives $y(t) = 0 \cdot e^{-P(t)} \equiv 0$, the trivial solution. By the uniqueness just proved it is the only solution with $y(t_0) = 0$. Equivalently: if a solution vanishes at one time t_1, apply the result with initial time t_1 to get $y \equiv 0$; so a non-trivial solution of this equation never vanishes (since $e^{-P} > 0$, it keeps the sign of y_0).

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check with $p(t) = 2t, t_0 = 0, y_0 = 3$: $P = t^2, y = 3e^{-t^2}, y' = -6t\,e^{-t^2} = -2t\,y$. Correct.

The same formula follows from the integrating-factor method (Problem 2.3): $\mu = e^P$, $(\mu y)' = 0$. The point of this exercise is that the argument above proves uniqueness directly, i.e. it does not presuppose the formula.

Interval of existence: the solution exists on the whole interval I where p is continuous, in agreement with the explicit formula, which is finite wherever p is continuous. This contrasts with non-linear equations such as Problem 2.13, where solutions can blow up in finite time.

RESULT

$$y(t) = y_0 \exp\left(-\int_{t_0}^{t} p(s)\,ds\right), \qquad y_0 = 0 \implies y \equiv 0.$$

EXERCISE 11

Invariance of Solution Ratios in Linear ODEs

Original statement · ProblemsCh2.pdf, p. 3

Linear homogeneous • Ratio is constant

PROBLEM STATEMENT

Without solving the ODE, show that if $y_1(t)$ and $y_2(t)$ are any two solutions of

$$\frac{dy}{dt} + p(t)\, y = 0,$$

then the ratio $y_1(t)/y_2(t)$ is constant.

Source: ProblemsCh2.pdf, Exercise 11 (page 3).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Assume p is continuous on an interval I, and y_1, y_2 are differentiable solutions on I: $y_i' = -p\, y_i$ for $i = 1, 2$. The ratio $r(t) = y_1(t)/y_2(t)$ is defined only where $y_2(t) \neq 0$. So we first have to settle where y_2 can vanish.

Fact needed: a solution of this equation either vanishes identically or has no zeros. This follows from uniqueness (Problem 2.10): if $y_2(t_1) = 0$ at some t_1, then y_2 and the zero function solve the same IVP at t_1, so $y_2 \equiv 0$. We assume $y_2 \not\equiv 0$ (otherwise the ratio is meaningless), hence $y_2(t) \neq 0$ on all of I.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: to show that a function is constant on an interval it is enough to show that its derivative is zero (Mean Value Theorem). The derivative of a quotient can be computed with the quotient rule, and the ODE lets us replace y_1' and y_2' by $-p\, y_1$ and $-p\, y_2$ without solving anything.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

By the formula of the notes with $q = 0$, every solution is $y = y(t_0)\, e^{-P(t)}$ with $P(t) = \int_{t_0}^{t} p$ (uniqueness from Problem 2.10), so $y_1/y_2 = y_1(t_0)/y_2(t_0)$ wherever $y_2(t_0) \neq 0$. The problem asks for the same conclusion without solving, which is the quotient-rule argument below.

PHASE 3: STEP-BY-STEP DERIVATION

Apply the quotient rule to $r = y_1/y_2$, on I where $y_2 \neq 0$:

$$r'(t) = \frac{y_1'(t)\,y_2(t) - y_1(t)\,y_2'(t)}{y_2(t)^2}.$$

Substitute $y_1' = -p\,y_1$ and $y_2' = -p\,y_2$:

$$r'(t) = \frac{(-p\,y_1)\,y_2 - y_1\,(-p\,y_2)}{y_2^2} = \frac{-p\,y_1y_2 + p\,y_1y_2}{y_2^2} = 0.$$

So $r' \equiv 0$ on the interval I, and by the Mean Value Theorem r is constant: $y_1(t) = c\,y_2(t)$ with

$$c = \frac{y_1(t_0)}{y_2(t_0)} \quad \text{for any } t_0 \in I.$$

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Consistency with the explicit solution (Problem 2.10): $y_i = y_i(t_0)\,e^{-P(t)}$ with $P(t) = \int_{t_0}^{t} p\,ds$, so $y_1/y_2 = y_1(t_0)/y_2(t_0)$, independent of t. Example: $p = 2t$, $y_1 = 3e^{-t^2}$, $y_2 = e^{-t^2}$, ratio 3.

Interpretation: the solution set of $y' + py = 0$ is a one-dimensional linear space, spanned by e^{-P}. Any two non-trivial solutions are proportional.

Limits: the result requires $y_2 \neq 0$ on the interval, which is guaranteed by the Fact in Phase 1. Without linearity (for example $y' = y^2$) the argument breaks down: the term $-p\,y_i$ is replaced by a non-proportional expression and r' does not vanish.

RESULT

$\dfrac{d}{dt}\left(\dfrac{y_1}{y_2}\right) = 0$, hence $y_1/y_2 \equiv c$ is constant.

EXERCISE 12

Trajectory Crossing and Uniqueness Bounds

Original statement · ProblemsCh2.pdf, p. 3

No crossing • Barrier solutions

PROBLEM STATEMENT

Consider $\dfrac{dy}{dt} = f(y,t)$, where f satisfies the hypothesis of the existence and uniqueness theorem. Suppose that

(i) $y_1(t) = -2$ for all t is a solution and we are studying a different solution for which $y(0) = 0$.

(ii) $y_1(t) = -t - 1$ and $y_2(t) = t^2 + 1$ are solutions and $y(0) = 0$.

Based on the uniqueness theorem, what can you conclude about the solutions in each case?

Source: ProblemsCh2.pdf, Exercise 12 (page 3).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Hypothesis of the theorem: f and $\partial f/\partial y$ are continuous on an open set $\Omega \subseteq \mathbb{R}^2$ containing the graphs of all the solutions considered. Then through every point $(t_*, y_*) \in \Omega$ passes exactly one solution (on its maximal interval of existence).

Consequence used below (no-crossing principle): if two solutions u, v of this ODE satisfy $u(t_*) = v(t_*)$ at some time t_* in the common interval, then they solve the same IVP at t_*, so $u \equiv v$ on the common interval. Contrapositive: two different solutions never meet. Hence $u - v$ has no zeros on the common interval.

The function f is not given explicitly, and we only use that the theorem applies. All conclusions are therefore restricted to the common interval of existence $J \ni 0$ of the solutions involved (a connected interval containing $t = 0$).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: the Intermediate Value Theorem states that a continuous function with no zeros on an interval has constant sign there. Apply it to $u - v$ for two solutions that never meet. The sign at $t = 0$, known from the initial data, then fixes the sign on all of J. This gives bounds without knowing f.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

Picard's theorem of the notes gives one solution per initial value, so two different solutions never meet: if they met at t_1, both would solve the IVP with data at t_1 and would coincide. This is the argument of step 1 of the logistic example of the notes (x never reaches 0 or M, so $M - x$ keeps the sign of $M - x_0$), used here with y_1 and y_2 in place of the constant solutions.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): $y_1 \equiv -2$ and a different solution y with $y(0) = 0$

y_1 is a solution and $y \not\equiv y_1$ (indeed $y(0) = 0 \neq -2 = y_1(0)$). Define $d(t) = y(t) - y_1(t) = y(t) + 2$. By the no-crossing principle, $d(t) \neq 0$ for all $t \in J$. d is continuous and $d(0) = 0 + 2 = 2 > 0$. By the Intermediate Value Theorem, a continuous function that changes sign must vanish somewhere between; since d never vanishes, it keeps the sign it has at $t = 0$:

$$y(t) + 2 > 0 \quad \Longleftrightarrow \quad y(t) > -2 \qquad \text{for all } t \in J.$$

Conclusion: the solution starting at $y(0) = 0$ stays strictly above the constant solution $y = -2$ for all time; it is bounded below by -2 and can never reach it.

Part (ii): $y_1 = -t - 1$, $y_2 = t^2 + 1$ and a solution y with $y(0) = 0$

First, check how the two given solutions are placed with respect to each other: $y_2(t) - y_1(t) = t^2 + t + 2$. Its discriminant is $1 - 8 = -7 < 0$, so $t^2 + t + 2 > 0$ for all t (its minimum is $7/4$ at $t = -\frac{1}{2}$). So $y_1 < y_2$ for all t, consistent with the no-crossing principle (they never meet).

Initial position of y: $y_1(0) = -1, y_2(0) = 1$, so $y_1(0) < y(0) = 0 < y_2(0)$. Apply the argument to each pair. $d_1 = y - y_1$ is continuous, $d_1(0) = 0 - (-1) = 1 > 0$ and never vanishes, hence $d_1 > 0$ on J. $d_2 = y_2 - y$ is continuous, $d_2(0) = 1 - 0 = 1 > 0$ and never vanishes, hence $d_2 > 0$ on J. Therefore

$$-t - 1 < y(t) < t^2 + 1 \qquad \text{for all } t \in J.$$

The solution is trapped between the two given solutions for as long as it exists.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Sanity check of the geometry: in (i), the line $y = -2$ and the curve $y(t)$ with $y(0) = 0$ are separated by a vertical gap 2 at $t = 0$; the gap is positive for all t. In (ii), the vertical gap between y_1 and y_2 at $t = 0$ is 2 and $y(0) = 0$ lies strictly inside; at $t = 1$ the bounds are $-2 < y(1) < 2$, and at $t = -1$ they are $0 < y(-1) < 2$.

Limits of the conclusion: it holds only on the common interval of existence J; the bounds alone do not say how large J is. Erratum: if f satisfies the hypotheses of the theorem on the whole region $\mathbb{R}^2$ (continuous with $\partial f/\partial y$ continuous), then in (ii) the solution cannot blow up in finite time, since it is trapped between y_1 and y_2, which are bounded on every bounded interval; hence y is defined for all $t \in \mathbb{R}$ (in (i) there is only a lower bound, so blow-up to $+\infty$ is not excluded). If f is only defined on a smaller region, J may be smaller. If f were not Lipschitz (for example $y' = 3y^{2/3}$, Problem 2.14), different solutions could touch or merge, and the bounds would be lost.

The data in (ii) are a consistency condition for f as well: two solutions that do not meet can coexist, whereas two that cross (say $y_1 = t$, $y_2 = -t$ intersecting at $t = 0$) would contradict the uniqueness hypothesis.

CONCLUSIONS

(i) $y(t) > -2$ for all t in the interval of existence. (ii) $-t - 1 < y(t) < t^2 + 1$ for all t in the interval of existence.

EXERCISE 13

Multi-Equilibria Autonomous Phase Line Dynamics

Original statement · ProblemsCh2.pdf, p. 3

Phase line • Finite-time blow-up

PROBLEM STATEMENT

Consider the autonomous differential equation

$$\frac{dy}{dt} = y(y-2)(y-3).$$

In each of the following cases, what can you conclude based on the uniqueness theorem about the different solutions if

(i) $y(0) = -1$,
(ii) $y(0) = 1$,
(iii) $y(0) = 2$,
(iv) $y(0) = 4$?

Source: ProblemsCh2.pdf, Exercise 13 (page 3).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Autonomous equation $\dot{y} = f(y)$ with the cubic polynomial

$$f(y) = y(y-2)(y-3) = y^3 - 5y^2 + 6y, \qquad f'(y) = 3y^2 - 10y + 6.$$

Both f and f' are continuous on all of $\mathbb{R}$, so the existence and uniqueness theorem holds at every initial point: through each $(0, y_0)$ passes exactly one solution, defined on a maximal open interval of existence.

Equilibria ($f(y^*) = 0$): $y^* = 0,\ 2,\ 3$. They give the constant solutions $y \equiv 0$, $y \equiv 2$, $y \equiv 3$.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: by uniqueness, two different solutions of the same autonomous equation never touch. A non-constant solution therefore cannot reach an equilibrium value in finite time, so it stays in one of the four intervals

$$I_1 = (-\infty, 0), \quad I_2 = (0, 2), \quad I_3 = (2, 3), \quad I_4 = (3, \infty)$$

determined by the equilibria. Inside an interval f has a constant sign, so $y(t)$ is strictly monotone. A monotone function that stays in a bounded interval has a limit, and this limit must be an equilibrium. If the interval is unbounded, the solution may escape to infinity; whether this happens in finite time is decided by separating variables.

Sign of f and stability from f':

Interval or point	**Sign of f (test point)**	**Motion**
$y < 0$	$f(-1) = (-1)(-3)(-4) = -12 < 0$	decreasing
$0 < y < 2$	$f(1) = (1)(-1)(-2) = 2 > 0$	increasing
$2 < y < 3$	$f(2.5) = (2.5)(0.5)(-0.5) = -0.625 < 0$	decreasing
$y > 3$	$f(4) = (4)(2)(1) = 8 > 0$	increasing

Stability: $f'(0) = 6 > 0$ (unstable), $f'(2) = 12 - 20 + 6 = -2 < 0$ (asymptotically stable), $f'(3) = 27 - 30 + 6 = 3 > 0$ (unstable). This agrees with the arrows: flow leaves 0 and 3 on both sides and enters 2 from both sides.

Time scale for blow-up: for $|y|$ large, $f(y) \sim y^3$, and the time needed to reach infinity is $\int^{\infty} dy/f(y) \sim \int^{\infty} dy/y^3 < \infty$. This finite integral signals that infinity is reached in finite time. We compute the time exactly by separation of variables.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The notes classify a stationary point x^* ($f(x^*) = 0$) by the sign of $f'(x^*)$: $f'(x^*) > 0$ unstable, $f'(x^*) < 0$ stable. With $f(y) = y(y-2)(y-3) = y^3 - 5y^2 + 6y$, $f'(y) = 3y^2 - 10y + 6$, which gives the three values above. The sketch argument of the notes (the solution moves right where $f > 0$ and left where $f < 0$) is the sign table, and both routes give the same classification.

PHASE 3: STEP-BY-STEP DERIVATION

Separation of variables with partial fractions

For $y \neq 0, 2, 3$ write $\dfrac{dy}{y(y-2)(y-3)} = dt$ and decompose:

$$\frac{1}{y(y-2)(y-3)} = \frac{A}{y} + \frac{B}{y-2} + \frac{C}{y-3}.$$

Cover-up rule: $A = \frac{1}{(0-2)(0-3)} = \frac{1}{6}$, $B = \frac{1}{2\,(2-3)} = -\frac{1}{2}$, $C = \frac{1}{3\,(3-2)} = \frac{1}{3}$. Check: $A + B + C = \frac{1}{6} - \frac{1}{2} + \frac{1}{3} = 0$, as required since the left side decays like y^{-3}. Integrating,

$$G(y) := \int \frac{dy}{y(y-2)(y-3)} = \frac{1}{6}\ln|y| - \frac{1}{2}\ln|y-2| + \frac{1}{3}\ln|y-3|.$$

Barrow's rule from $(0, y_0)$: $G(y(t)) - G(y_0) = t$. Equivalently, multiplying by 6 and exponentiating,

$$\frac{|y|\,|y-3|^2}{|y-2|^3} = K\,e^{6t}, \qquad K = \frac{|y_0|\,|y_0-3|^2}{|y_0-2|^3}.$$

Limit of G at infinity: the coefficients sum to zero, so $G(y) = \frac{1}{6}\ln\dfrac{|y|\,|y-3|^2}{|y-2|^3}$, and the argument of the logarithm tends to 1 as $y \to \pm\infty$. Hence $G(\pm\infty) = \frac{1}{6}\ln 1 = 0$.

Case (i): $y(0) = -1$

$y(0) \in I_1$. By uniqueness the solution cannot reach the equilibrium $y = 0$, so $y(t) < 0$ for all t in its domain. Since $f < 0$ there, y is strictly decreasing.

- Backward in time: y increases and is bounded above by 0, so it tends to an equilibrium: $y \to 0^-$ as $t \to -\infty$.
- Forward in time: y decreases without a lower barrier and $f \sim y^3$, so it blows up to $-\infty$ at a finite time T^*.

Blow-up time from Barrow's rule, $T^* = G(-\infty) - G(-1) = -G(-1)$:

$$G(-1) = \frac{1}{6}\ln 1 - \frac{1}{2}\ln 3 + \frac{1}{3}\ln 4, \qquad T^* = \frac{1}{2}\ln 3 - \frac{1}{3}\ln 4 = \frac{1}{2}\ln 3 - \frac{2}{3}\ln 2 \approx 0.0872.$$

Cross-check with the closed form: $K = \frac{1\cdot 16}{27}$, and blow-up occurs when the left side tends to 1 (as $|y| \to \infty$): $1 = \frac{16}{27}e^{6T^*}$, so $T^* = \frac{1}{6}\ln\frac{27}{16} = \frac{1}{6}(3\ln 3 - 4\ln 2)$, the same value.

Case (ii): $y(0) = 1$

$y(0) \in I_2 = (0, 2)$. The equilibria 0 and 2 are barriers, so $0 < y(t) < 2$ for all t. A bounded solution cannot blow up, so it is defined for all $t \in \mathbb{R}$. Since $f > 0$, y is strictly increasing, hence it has limits at both ends and they are equilibria:

$$\lim_{t\to-\infty} y(t) = 0, \qquad \lim_{t\to+\infty} y(t) = 2.$$

The solution goes from the unstable equilibrium 0 to the stable equilibrium 2 (a monotone, sigmoidal profile). Neither limit is reached in finite time.

Case (iii): $y(0) = 2$

$f(2) = 0$, so $y(t) \equiv 2$ is a solution of the IVP. By the uniqueness theorem it is the only one:

$$y(t) \equiv 2 \quad \forall t \in \mathbb{R}.$$

In particular no non-constant solution can pass through the point $(t, 2)$ at any time; solutions that start in I_2 or I_3 approach 2 only asymptotically.

Case (iv): $y(0) = 4$

$y(0) \in I_4 = (3, \infty)$. The barrier $y = 3$ gives $y(t) > 3$ for all t; $f > 0$ there, so y is strictly increasing.

- Backward in time: bounded below by 3 and decreasing as t decreases, so $y \to 3^+$ as $t \to -\infty$.
- Forward in time: no upper barrier and $f \sim y^3$, so $y \to +\infty$ at a finite time T^*.

$$T^* = G(+\infty) - G(4) = -G(4) = -\left[\frac{1}{6}\ln 4 - \frac{1}{2}\ln 2 + \frac{1}{3}\ln 1\right] = -\left[\frac{1}{3}\ln 2 - \frac{1}{2}\ln 2\right] = \frac{1}{6}\ln 2 \approx 0.1155.$$

Cross-check with the closed form: $K = \frac{4\cdot 1}{8} = \frac{1}{2}$, and $1 = \frac{1}{2}e^{6T^*}$ gives $T^* = \frac{1}{6}\ln 2$.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Numerical checks (computed): the partial-fraction decomposition from sympy is $\frac{1}{6y} - \frac{1}{2(y-2)} + \frac{1}{3(y-3)}$, matching A, B, C. An independent fourth-order Runge-Kutta integration of $\dot y = y(y-2)(y-3)$ with step 1.5×10^{-6} reaches $|y| > 10^7$ at $t \approx 0.115526$ for $y_0 = 4$ (formula: 0.115525) and at $t \approx 0.087210$ for $y_0 = -1$ (formula: 0.087208).

Time scale: the correct statement for $y_0 = -1$ and $y_0 = 4$ is not $y \to \pm\infty$ as $t \to \infty$; the solutions cease to exist at $T^* \approx 0.087$ and $T^* \approx 0.116$ respectively, so a limit $t \to \infty$ is meaningless for them. By contrast, for $y_0 = 1$ the solution is global and the limit $t \to \infty$ is 2.

Dimensional remark: if y is a non-dimensional state and t is measured in units of a characteristic time τ, then T^* is a number of such units; the equation $\dot y = f(y)$ requires $[f] = [y]/[t]$, and the coefficients $2, 3$ carry the dimensions of y.

CLASSIFICATION OF TRAJECTORIES

Initial condition	Region	Behavior forward	Behavior backward
(i) $y(0) = -1$	$y < 0$, decreasing	blow-up to $-\infty$ at $T^* = \frac{1}{2}\ln 3 - \frac{2}{3}\ln 2 \approx 0.087$	$y \to 0^-$ as $t \to -\infty$
(ii) $y(0) = 1$	$0 < y < 2$, increasing	$y \to 2$ as $t \to \infty$	$y \to 0^+$ as $t \to -\infty$
(iii) $y(0) = 2$	equilibrium	$y \equiv 2$	$y \equiv 2$
(iv) $y(0) = 4$	$y > 3$, increasing	blow-up to $+\infty$ at $T^* = \frac{1}{6}\ln 2 \approx 0.116$	$y \to 3^+$ as $t \to -\infty$

EXERCISE 14

Non-Lipschitz Branching and Picard's Theorem

Original statement · ProblemsCh2.pdf, p. 4

Non-uniqueness • Failure of Lipschitz

PROBLEM STATEMENT

Given the IVP

$$\frac{dy}{dt} = 3y^{2/3}, \qquad y(0) = 0,$$

on the rectangle $R = \{(y,t) \in \mathbb{R}^2 : |y| \le 1,\ |t| \le 1\}$. Show that both $y_1(t) = t^3$ and $y_2(t) = 0$ are solutions. Does this contradict Picard's theorem?

Source: ProblemsCh2.pdf, Exercise 14 (page 4).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Right-hand side $f(y,t) = 3y^{2/3}$, where $y^{2/3} = (y^{1/3})^2 = |y|^{2/3} \ge 0$ is defined for all real y (real cube root). It is continuous on R. Its partial derivative is

$$\frac{\partial f}{\partial y} = 2\,y^{-1/3} \quad (y \neq 0),$$

which is unbounded as $y \to 0$ and not defined at $y = 0$, which is precisely the initial value $y(0) = 0$. So the hypotheses of Picard's theorem (continuity of f and $\partial f/\partial y$ near the initial point, or a Lipschitz condition in y) are not satisfied at $(0,0)$.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: to show that a candidate is a solution of an IVP we verify the ODE and the initial condition by direct substitution. To decide whether the non-uniqueness contradicts Picard's theorem, we test the Lipschitz condition $|f(y,t) - f(\tilde{y},t)| \le L|y-\tilde{y}|$ near $y=0$; if it fails, the theorem does not apply and no contradiction arises.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

In the form of the notes, $y' = f(t,y)$ with $f = 3y^{2/3}$, Picard's theorem needs f and $\partial f/\partial y$ continuous near $(t_0,y_0) = (0,0)$. Here f is continuous, but $\frac{\partial f}{\partial y} = 3 \cdot \frac{2}{3}\, y^{-1/3} = 2y^{-1/3}$ is not defined at $y=0$ and is unbounded as $y \to 0$. The hypothesis fails, so the theorem says nothing at $(0,0)$; the Lipschitz test below (Robinson's form of the hypothesis) reaches the same conclusion.

PHASE 3: STEP-BY-STEP DERIVATION

Verification of $y_1(t) = t^3$

Initial condition: $y_1(0) = 0$. Derivative: $y_1'(t) = 3t^2$. Right-hand side with the real cube root: $y_1^{1/3} = (t^3)^{1/3} = t$, so $y_1^{2/3} = t^2$ and $3y_1^{2/3} = 3t^2$. Both sides agree for all t. The graph stays in R for $|t| \le 1$, since $|y_1| = |t|^3 \le 1$.

Verification of $y_2(t) = 0$

$y_2' = 0$ and $3 \cdot 0^{2/3} = 0$, and $y_2(0) = 0$. It is a solution, and its graph is inside R.

Two distinct solutions, $y_1(t) = t^3$ and $y_2(t) = 0$, of the same IVP exist on $[-1,1]$.

Failure of the Lipschitz condition

Take $\tilde{y} = 0$ and $y > 0$:

$$\frac{|f(y,t) - f(0,t)|}{|y-0|} = \frac{3y^{2/3}}{y} = 3\,y^{-1/3} \longrightarrow +\infty \quad (y \to 0^+).$$

No constant L bounds this quotient, so f is not Lipschitz in y on any neighborhood of $y=0$ in R. Consistently, $\partial f/\partial y = 2y^{-1/3}$ is unbounded.

Where the branching comes from

For $y > 0$ the equation is separable: $\dfrac{dy}{3y^{2/3}} = dt$. Integrating, $y^{1/3} = t + c$, so $y = (t+c)^3$ for $t + c > 0$. For every $c \in [0,1]$ define

$$y_c(t) = \begin{cases} 0, & -1 \le t \le c, \\ (t-c)^3, & c < t \le 1. \end{cases}$$

At $t = c$ the one-sided derivatives are both 0 ($\frac{d}{dt}(t-c)^3 = 3(t-c)^2 \to 0$), so y_c is C^1, it satisfies the ODE on each piece and $y_c(0) = 0$. Hence the IVP has infinitely many solutions, and $y_2 = 0$ (the limit $c \ge 1$) is a member of the family. Note that y_c with $c = 0$ is 0 for $t \le 0$ and t^3 for $t > 0$; the function $y_1 = t^3$ on all of $[-1, 1]$ is a different solution, obtained by using the negative-side branch $y = (t - c)^3, t < c$, with $c = 0$ (valid since $y^{1/3} = t - c$ requires only y and $t - c$ to have the same sign, and $\frac{d}{dt}t^3 = 3t^2 = 3(t^3)^{2/3}$).

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Answer to the question: no contradiction. Picard's theorem (Robinson, Theorem 6.2) guarantees uniqueness only if f and $\partial f/\partial y$ are continuous (or f is Lipschitz in y) near the initial point. Here f is continuous but $\partial f/\partial y = 2y^{-1/3}$ is not defined at $y = 0$ and the Lipschitz quotient is unbounded, so the hypotheses fail exactly at the initial point $(0, 0)$. The theorem is silent in that case, and non-uniqueness is possible.

Contrast: for any initial value $y_0 \neq 0$ with $t_0 \in (-1, 1)$, f and $\partial f/\partial y$ are continuous in a neighborhood and the solution is unique; the branching occurs only from the equilibrium $y = 0$ (compare Problem 2.9 (iii) and (iv)).

Why the solution can leave $y = 0$: the time to travel from 0 to y is $\int_0^y ds/(3s^{2/3}) = y^{1/3}$, which is finite and tends to 0 as $y \to 0$. For Lipschitz right-hand sides this integral diverges (compare $f = y$, where $\int_0 ds/s = \infty$), so an equilibrium cannot be left in finite time.

RESULT

$y_1 = t^3$ and $y_2 = 0$ both solve $\dot{y} = 3y^{2/3}, y(0) = 0$. No contradiction with Picard's theorem: its hypotheses fail at $(0, 0)$ because $\partial f/\partial y = 2y^{-1/3}$ is unbounded.

EXERCISE 15

Singular ODE and Domain of Definition

Original statement · ProblemsCh2.pdf, p. 4

Singular coefficient • Non-uniqueness at t = 0

PROBLEM STATEMENT

Consider the differential equation

$$\frac{dy}{dt} = \frac{2y+1}{t}.$$

Compute the general solution and show that there are two solutions such that $y(0) = -1/2$. Does this contradict the uniqueness theorem?

Source: ProblemsCh2.pdf, Exercise 15 (page 4).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Right-hand side $f(y,t) = (2y+1)/t$. It is continuous, together with $\partial f/\partial y = 2/t$, on each of the half-planes $t > 0$ and $t < 0$, but it is not defined on the line $t = 0$. So the equation is singular at $t = 0$; any solution of the differential equation in the usual sense is defined on $t > 0$ or on $t < 0$ separately. On each half-plane Theorem 6.2 gives a unique solution through every point.

Equilibrium on $t \neq 0$: $2y + 1 = 0$, i.e. $y \equiv -\frac{1}{2}$.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: the equation is separable ($g(t) = 1/t$, $h(y) = 2y+1$) and also linear. We solve it by separation of variables on each half-line $t \neq 0$, and use the integrating-factor method as an independent check. Then we study what happens at $t = 0$, where the hypotheses of the uniqueness theorem fail.

Change of variables written explicitly: $u = 2y + 1$, $du = 2\,dy$, which reduces the equation to $du/dt = 2u/t$.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The equation is separable in the form of the notes, $y' = f(y)\,g(t)$ with $f(y) = 2y + 1$ and $g(t) = 1/t$: if $f \neq 0$ (never, since $y \equiv -1/2$ is the constant solution), divide by f and integrate from t_0 to t on a half-line that does not contain 0.

Picard's theorem of the notes needs the right side $F(t, y) = \frac{2y+1}{t}$ and $\frac{\partial F}{\partial y} = \frac{2}{t}$ continuous near (t_0, y_0). This holds for $t_0 \neq 0$ and fails at $t_0 = 0$, which is exactly where the solutions with $y(0) = -1/2$ are not unique.

PHASE 3: STEP-BY-STEP DERIVATION

General solution by separation of variables

Equilibrium $y \equiv -\frac{1}{2}$. For $y \neq -\frac{1}{2}$ and $t \neq 0$, separate:

$$\frac{dy}{2y+1} = \frac{dt}{t}.$$

Integrate, with $u = 2y + 1$, $dy = du/2$ on the left:

$$\frac{1}{2}\ln|2y+1| = \ln|t| + c \quad \Longrightarrow \quad |2y+1| = e^{2c}\,t^2.$$

Let $C = \pm e^{2c} \in \mathbb{R} \setminus \{0\}$. Then $2y + 1 = C\,t^2$. The equilibrium corresponds to $C = 0$, so

$$y(t) = -\frac{1}{2} + \frac{C}{2}\,t^2 = -\frac{1}{2} + k\,t^2, \qquad k \in \mathbb{R}.$$

Independent check with an integrating factor on $t > 0$: $y' - \frac{2}{t}y = \frac{1}{t}$, $\mu = e^{-2\ln t} = t^{-2}$, $(t^{-2}y)' = t^{-3}$, so $t^{-2}y = -\frac{1}{2}t^{-2} + k$ and $y = -\frac{1}{2} + k\,t^2$, the same family. On $t < 0$ the same formula holds with $\mu = t^{-2}$ again. The constants k on $t > 0$ and on $t < 0$ are independent because the two half-lines are disconnected.

Solutions through $y(0) = -\frac{1}{2}$

Every member of the family has the limit $y(t) \to -\frac{1}{2} + 0 = -\frac{1}{2}$ as $t \to 0$. So each of them extends continuously to $t = 0$ with $y(0) = -\frac{1}{2}$, whatever k is. In particular, for $k = 0$ and $k = 1$:

$$y_a(t) = -\frac{1}{2}, \qquad y_b(t) = -\frac{1}{2} + t^2, \qquad \text{both satisfy } y(0) = -\frac{1}{2}.$$

Both are smooth on all of $\mathbb{R}$ and satisfy the equation in the form $t\,y' = 2y + 1$ at $t = 0$ as well (both sides vanish there: $0 = 0$). Hence there are two (in fact infinitely many, one for each k, and even different constants k_+, k_- for $t > 0$ and $t < 0$, which still give a C^1 function at $t = 0$) solutions of the IVP.

Conversely, no solution can have $y(0) = y_0 \neq -\frac{1}{2}$: from $2y + 1 = Ct^2$ we get $y \to -\frac{1}{2}$ as $t \to 0$. So the initial value problem at $t = 0$ is solvable only for $y_0 = -\frac{1}{2}$, and then it has infinitely many solutions.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Substitution check: for $y = -\frac{1}{2} + k\,t^2$, $y' = 2kt$, and $\frac{2y+1}{t} = \frac{2kt^2}{t} = 2kt$. Equal for $t \neq 0$.

Does this contradict the uniqueness theorem? No. The theorem requires $f(y, t)$ and $\partial f/\partial y$ to be continuous in a neighborhood of the initial point $(t_0, y_0) = (0, -\frac{1}{2})$. Here $f = (2y + 1)/t$ is not even defined at $t = 0$, and $\partial f/\partial y = 2/t$ is unbounded as $t \to 0$; no neighborhood of $(0, -\frac{1}{2})$ lies inside a region where the hypotheses hold. The multiple solutions all pass through the singular point of the equation. For any initial time $t_0 \neq 0$ the solution is unique, in agreement with the theorem.

Domain of definition: a genuine solution of $y' = (2y + 1)/t$ lives on $t > 0$ or $t < 0$. The functions above are solutions of the equation on each half-line and extend to $t = 0$ as solutions of the equivalent form $t\,y' = 2y + 1$, whose coefficient of y' vanishes at $t = 0$; this is why uniqueness is lost there.

RESULT

General solution $y = -\frac{1}{2} + k\,t^2$. Through $y(0) = -\frac{1}{2}$ pass infinitely many solutions (for instance $y = -\frac{1}{2}$ and $y = -\frac{1}{2} + t^2$). No contradiction with the uniqueness theorem, because f is singular at $t = 0$.

EXERCISE 16

Pitchfork Phase Line and Stability Regimes

Original statement · ProblemsCh2.pdf, p. 4

Phase line • Stability • Explicit solution

PROBLEM STATEMENT

Consider the autonomous equation

$$\frac{dx}{dt} = x\left(\kappa^2 - x^2\right)$$

with a general initial condition $x(0) = x_0$.

(i) Calculate the stationary solutions, which satisfy $dx/dt = 0$;
(ii) Calculate and sketch the solutions for $t > 0$, discussing the different behaviors which can be obtained in terms of the relative values of $\kappa > 0$ and x_0, with the help of the uniqueness theorem.

Source: ProblemsCh2.pdf, Exercise 16 (page 4).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Autonomous equation $\dot{x} = f(x)$ with $f(x) = x(\kappa^2 - x^2) = \kappa^2 x - x^3$ and parameter $\kappa > 0$. We have

$$f'(x) = \kappa^2 - 3x^2.$$

Both f and f' are polynomials, hence continuous on $\mathbb{R}$: the IVP has a unique solution for every $x_0 \in \mathbb{R}$, on a maximal interval of existence. The right-hand side is odd, $f(-x) = -f(x)$, so if $x(t)$ is a solution then $-x(t)$ is also a solution (symmetry $x \to -x$).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why these methods: (a) the phase line (sign of f between equilibria) and the no-crossing consequence of uniqueness give the qualitative picture for every x_0 without solving; (b) the equation is a Bernoulli equation ($\alpha = 3$, see Problem 2.8), so the substitution $w = x^{-2}$ turns it into a linear equation and gives the exact solution to confirm the qualitative picture and to find timescales.

Change of variables written explicitly: $w = x^{1-\alpha} = x^{-2}$ (valid for $x \neq 0$), $w' = -2x^{-3}x'$, $w(0) = 1/x_0^2$.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The test of the notes: a stationary point x^* with $f'(x^*) > 0$ is unstable, with $f'(x^*) < 0$ stable. Here $f(x) = \kappa^2 x - x^3$ and $f'(x) = \kappa^2 - 3x^2$: $f'(0) = \kappa^2 > 0$ (unstable) and $f'(\pm\kappa) = \kappa^2 - 3\kappa^2 = -2\kappa^2 < 0$ (stable). Between consecutive zeros the sign of f fixes the direction of motion, as in the sketch of the notes, and Picard's theorem (f and f' continuous everywhere) makes the stationary solutions barriers that no other solution crosses.

PHASE 3: STEP-BY-STEP DERIVATION

Part (i): stationary solutions

Solve $f(x^*) = x^*(\kappa - x^*)(\kappa + x^*) = 0$ using $\kappa^2 - x^2 = (\kappa - x)(\kappa + x)$:

$$x^* = 0, \qquad x^* = \kappa, \qquad x^* = -\kappa.$$

Stability from $f'(x^*)$: $f'(0) = \kappa^2 > 0$ (unstable); $f'(\pm\kappa) = \kappa^2 - 3\kappa^2 = -2\kappa^2 < 0$ (asymptotically stable). Therefore the constant solutions are $x \equiv 0$ (unstable) and $x \equiv \pm\kappa$ (stable).

Part (ii), step 1: phase line from uniqueness

By uniqueness solutions cannot cross the equilibria, so the equilibria $-\kappa, 0, \kappa$ split $\mathbb{R}$ into four invariant intervals. Sign of $f = x(\kappa - x)(\kappa + x)$:

Region	**Signs of $x, \kappa - x, \kappa + x$**	**Sign of f**	**Motion**
$x > \kappa$	$+, -, +$	$-$	decreases to κ
$0 < x < \kappa$	$+, +, +$	$+$	increases to κ
$-\kappa < x < 0$	$-, +, +$	$-$	decreases to $-\kappa$
$x < -\kappa$	$-, +, -$	$+$	increases to $-\kappa$

In each region the solution is monotone and bounded by the neighboring equilibria, hence global in time, with a limit that is an equilibrium: all solutions with $x_0 > 0$ tend to κ, all with $x_0 < 0$ tend to $-\kappa$, and $x_0 = 0$ stays at 0.

Part (ii), step 2: exact solution

Let $x_0 \neq 0$ (then $x(t) \neq 0$ for all t, by uniqueness). Compute $w = x^{-2}$ using the chain rule:

$$w' = -2x^{-3}x' = -2x^{-3}\left(\kappa^2 x - x^3\right) = -2\kappa^2 x^{-2} + 2 = -2\kappa^2 w + 2.$$

Linear equation $w' + 2\kappa^2 w = 2$. Integrating factor: $\int 2\kappa^2\,dt = 2\kappa^2 t$, $\mu = e^{2\kappa^2 t}$, $\mu(0) = 1$, and $(\mu w)' = 2e^{2\kappa^2 t}$. Barrow from 0 to t:

$$w(t)\,e^{2\kappa^2 t} - w(0) = \int_0^t 2e^{2\kappa^2 s}\,ds = \frac{1}{\kappa^2}\left[e^{2\kappa^2 s}\right]_0^t = \frac{1}{\kappa^2}\left(e^{2\kappa^2 t} - 1\right).$$

Hence, with $w(0) = 1/x_0^2$,

$$w(t) = \frac{1}{\kappa^2} + \left(\frac{1}{x_0^2} - \frac{1}{\kappa^2}\right)e^{-2\kappa^2 t} = \frac{x_0^2 + (\kappa^2 - x_0^2)\,e^{-2\kappa^2 t}}{\kappa^2 x_0^2}.$$

Since $x = \pm w^{-1/2}$ with the sign fixed by x_0 (a solution cannot change sign), and $x(0) = x_0$:

$$x(t) = \frac{\kappa\, x_0}{\sqrt{x_0^2 + (\kappa^2 - x_0^2)\,e^{-2\kappa^2 t}}}, \qquad x_0 \neq 0.$$

(For $x_0 = 0$ the solution is $x \equiv 0$, which the formula does not cover.) Check: at $t = 0$, the square root is $\sqrt{\kappa^2} = \kappa$, so $x(0) = x_0$.

Part (ii), step 3: behavior for $t > 0$

Write $E(t) = e^{-2\kappa^2 t} \in (0, 1]$ for $t \geq 0$. The radicand is $x_0^2(1 - E) + \kappa^2 E$, a convex combination of $x_0^2 > 0$ and $\kappa^2 > 0$, hence positive for all $t \geq 0$: the solution exists for all $t > 0$. As $t \to \infty$, $E \to 0$, the radicand $\to x_0^2$ and

$$x(t) \to \frac{\kappa\, x_0}{|x_0|} = \kappa\ \mathrm{sgn}(x_0).$$

Initial condition	**Behavior for $t > 0$**	**Limit as $t \to \infty$**
$x_0 > \kappa$	decreasing, stays above κ	κ
$x_0 = \kappa$	constant $x \equiv \kappa$	κ
$0 < x_0 < \kappa$	increasing, stays below κ	κ

Initial condition	**Behavior for $t > 0$**	**Limit as $t \to \infty$**
$x_0 = 0$	constant $x \equiv 0$	0 (unstable)
$-\kappa < x_0 < 0$	decreasing, stays above $-\kappa$	$-\kappa$
$x_0 = -\kappa$	constant $x \equiv -\kappa$	$-\kappa$
$x_0 < -\kappa$	increasing, stays below $-\kappa$	$-\kappa$

Shape of the sketch: for $0 < x_0 < \kappa$ the curve is sigmoidal if $x_0 < \kappa/\sqrt{3}$ (inflection point where $x'' = f'(x)\,x' = 0$, i.e. $f'(x) = \kappa^2 - 3x^2 = 0$, at $x = \kappa/\sqrt{3}$, where the slope is maximal), and concave without inflection if $x_0 \geq \kappa/\sqrt{3}$. The curves for $x_0 < 0$ are mirror images ($x \to -x$).

Behavior backward in time (for completeness): the radicand vanishes when $E = x_0^2/(x_0^2 - \kappa^2)$, which requires $|x_0| > \kappa$ and happens at $t = -\frac{1}{2\kappa^2} \ln \frac{x_0^2}{x_0^2 - \kappa^2} < 0$. So solutions with $|x_0| > \kappa$ blow up in finite negative time, while those with $0 < |x_0| < \kappa$ exist for all $t < 0$ and tend to 0 as $t \to -\infty$.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Check the ODE by differentiating the explicit solution. With $D(t) = x_0^2 + (\kappa^2 - x_0^2)e^{-2\kappa^2 t}$: $x = \kappa x_0 D^{-1/2}$, $x' = -\frac{1}{2}\kappa x_0 D^{-3/2} D'$, and $D' = -2\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}$. Also $\kappa^2 - x^2 = \kappa^2 - \frac{\kappa^2 x_0^2}{D} = \frac{\kappa^2(D - x_0^2)}{D} = \frac{\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}}{D}$, so $x(\kappa^2 - x^2) = \kappa x_0 D^{-1/2} \cdot \frac{\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}}{D}$. This equals x': $-\frac{1}{2}\kappa x_0 D^{-3/2} \cdot \left(-2\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}\right) = \kappa x_0 D^{-3/2}\kappa^2(\kappa^2 - x_0^2)e^{-2\kappa^2 t}$. They agree.

Numerical check (computed): for $\kappa = 1$, a fourth-order Runge-Kutta integration over $t \in [0, 5]$ (20000 steps) matches the closed form to better than 10^{-14} for $x_0 = 0.3,\ 2,\ -0.5,\ -3$. At $t = 5$ the distance to $\kappa = 1$ is 2.3×10^{-4} for $x_0 = 0.3$ and 1.7×10^{-5} for $x_0 = 2$, consistent with the time constant $1/(2\kappa^2) = 0.5$.

Time scale and dimensions: the linearization at $x = \pm\kappa$ has rate $f'(\pm\kappa) = -2\kappa^2$, so the approach to $\pm\kappa$ is exponential with time constant $1/(2\kappa^2)$ (visible in the factor $e^{-2\kappa^2 t}$), and at $x = 0$ the repulsion rate is κ^2. The equation is written in non-dimensional form: κ^2 is a rate with units $[t]^{-1}$, and x is measured in units for which the cubic coefficient equals 1.

Bifurcation context (Robinson, Section 7.6): for the parameter $\mu = \kappa^2$ the equation $\dot{x} = \mu x - x^3$ has one equilibrium $x = 0$ (stable) for $\mu < 0$, and for $\mu > 0$ the origin becomes unstable and two stable branches $\pm\sqrt{\mu}$ appear: a supercritical pitchfork. Here $\kappa > 0$ is the case $\mu > 0$.

RESULT

Equilibria 0 (unstable) and $\pm\kappa$ (stable). For $x_0 \neq 0$: $x(t) = \kappa x_0 / \sqrt{x_0^2 + (\kappa^2 - x_0^2)e^{-2\kappa^2 t}} \to \kappa\ \mathrm{sgn}(x_0)$ as $t \to \infty$.

EXERCISE 17

Direct Difference Method for Uniqueness

Original statement · ProblemsCh2.pdf, p. 4

Existence by inspection • Uniqueness by difference

PROBLEM STATEMENT

Show existence and uniqueness for the initial value problem (IVP)

$$\frac{dy}{dt} + a\,y = 0, \qquad y(t_0) = y_0.$$

For existence, just find a solution by inspection. Then suppose that there are two different solutions and argue that they have to be the same by taking their difference.

Source: ProblemsCh2.pdf, Exercise 17 (page 4).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Data: a constant $a \in \mathbb{R}$ (of either sign), an initial time t_0 and an initial value $y_0 \in \mathbb{R}$. A solution is a differentiable function $y : \mathbb{R} \to \mathbb{R}$ with $y' + ay = 0$ and $y(t_0) = y_0$. This is a special case of $y' + p(t)y = 0$ with $p \equiv a$ continuous, so the result also follows from Problem 2.10 or from Picard's theorem; here we are asked for a direct proof.

We prove two statements: (E) a solution exists, (U) two solutions of the same IVP coincide. Tools allowed: differentiation, linearity of the derivative, and the fact that a function with zero derivative on an interval is constant (Mean Value Theorem).

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Existence by inspection: the exponential function reproduces itself under differentiation up to a constant factor, which is exactly the behavior required by $y' = -a\,y$. We propose $y(t) = y_0 e^{-a(t-t_0)}$ and verify it.

Uniqueness by the difference: since the equation is linear and homogeneous, the difference of two solutions solves the same equation with zero initial data. So it is enough to prove that the only solution of $w' + aw = 0$, $w(t_0) = 0$, is $w \equiv 0$. Two routes are given.

- Approach A (squared difference): $E = w^2 \geq 0$ satisfies $E' = -2a\,E$, and $\left(E\,e^{2a(t-t_0)}\right)' = 0$.
- Approach B (integrating factor): $\left(w\,e^{a(t-t_0)}\right)' = 0$ directly.

In both approaches we must not write down the solution formula of the equation for w or E, because that formula presupposes the uniqueness we are proving. We only differentiate a product and apply the Mean Value Theorem.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The three-step algorithm of the notes with $p = a$, $q = 0$ gives the candidate: $P(t) = \int_{t_0}^{t} a\, d\tau = a(t - t_0)$, $I(t) = e^{a(t-t_0)}$, $I(t_0) = 1$, and $y(t) = \frac{y_0}{I(t)} = y_0\, e^{-a(t-t_0)}$. Approach B below is the uniqueness half of that algorithm: Step 2 with $q = 0$ says $(Iw)' = 0$.

Picard's theorem of the notes would also give uniqueness ($f = -a\,y$ and $\partial f/\partial y = -a$ are continuous), but only on a small interval $|t - t_0| < h$; the difference argument gives it on all of $\mathbb{R}$.

PHASE 3: STEP-BY-STEP DERIVATION

Existence

Candidate: $y(t) = y_0\, e^{-a(t-t_0)}$. Initial condition: $y(t_0) = y_0 e^0 = y_0$. Differentiate with the chain rule, $\frac{d}{dt}\left[-a(t - t_0)\right] = -a$:

$$y'(t) = y_0\, e^{-a(t-t_0)} \cdot (-a) = -a\, y(t) \quad \Longrightarrow \quad y'(t) + a\, y(t) = 0 \quad \forall t \in \mathbb{R}.$$

So y is a solution and existence is proved.

Uniqueness: the difference

Let y_1, y_2 be two solutions of the same IVP and define $w = y_1 - y_2$. Then $w(t_0) = y_0 - y_0 = 0$, and by linearity of the derivative

$$w' + a\, w = (y_1' + a y_1) - (y_2' + a y_2) = 0 - 0 = 0.$$

Approach A: squared difference

Let $E(t) = w(t)^2 \geq 0$. By the chain rule, $E' = 2\, w\, w'$, and using $w' = -a\, w$:

$$E' = 2w\,(-a\, w) = -2a\, w^2 = -2a\, E, \qquad \text{i.e.} \quad E' + 2a\, E = 0.$$

Multiply by $e^{2a(t-t_0)}$ and use the product rule:

$$\frac{d}{dt}\left[E(t)\,e^{2a(t-t_0)}\right] = \left(E' + 2a\,E\right)e^{2a(t-t_0)} = 0.$$

By the Mean Value Theorem, $E(t)\,e^{2a(t-t_0)}$ is constant, equal to its value at t_0, which is $E(t_0)\,e^0 = w(t_0)^2 = 0$. Since $e^{2a(t-t_0)} > 0$, we can divide:

$$E(t) = 0 \quad \forall t \in \mathbb{R}, \qquad \text{so} \quad w(t)^2 = 0 \implies w(t) = 0.$$

This holds for either sign of a. The shorter argument 'E decreases and is non-negative, hence zero' would only work for $a \geq 0$ and forward in time, which is why the product $E\,e^{2a(t-t_0)}$ is used.

Approach B: integrating factor on w

Multiply $w' + aw = 0$ by $e^{a(t-t_0)}$:

$$\frac{d}{dt}\left[w(t)\,e^{a(t-t_0)}\right] = \left(w' + a\,w\right)e^{a(t-t_0)} = 0.$$

By the Mean Value Theorem $w(t)\,e^{a(t-t_0)} = w(t_0)\,e^0 = 0$, and since the exponential never vanishes, $w(t) = 0$ for all t.

In both approaches $w \equiv 0$, i.e. $y_1 \equiv y_2$. Uniqueness is proved.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Numerical sanity check: $a = 2, t_0 = 1, y_0 = 3$: $y(t) = 3e^{-2(t-1)}$, $y(1) = 3$, $y'(t) = -6e^{-2(t-1)} = -2y$. For $a = -1.5$ the same formula gives exponential growth, and the uniqueness argument is unchanged.

Why the restriction to the difference is legitimate: linearity. For a non-linear equation, the difference of two solutions would not solve the same equation and more work (a Lipschitz estimate or Gronwall's inequality) would be needed, which is the content of Picard's theorem.

Relation to other exercises: Approach B is exactly the proof of Problem 2.10 with $p \equiv a$ and $y_0 = 0$; Approach A uses the quantity w^2, the one-dimensional analogue of an energy functional.

RESULT

Existence: $y(t) = y_0\,e^{-a(t-t_0)}$. Uniqueness: the difference w of two solutions satisfies $w' + aw = 0$, $w(t_0) = 0$, and both $\left(w^2 e^{2a(t-t_0)}\right)' = 0$ and $\left(w\,e^{a(t-t_0)}\right)' = 0$ force $w \equiv 0$.